ADVANCED ENERGY MATERIALS
www.advenergymat.de

Toward Efficiency Limits of Crystalline Silicon Solar Cells: Recent Progress in High-Efficiency Silicon Heterojunction Solar Cells

*Zhaoqing Sun, Xiaoqing Chen, Yongcai He, Jingjie Li, Jianqiang Wang, Hui Yan, and Yongzhe Zhang**

Photovoltaic (PV) technology is ready to become one of the main energy sources of, and contributers to, carbon neutrality by the mid-21st century. In 2020, a total of 135 GW of PV modules were produced. Crystalline silicon solar cells dominate the world's PV market due to high power conversion efficiency, high stability, and low cost. Silicon heterojunction (SHJ) solar cells are one of the promising technologies for next-generation crystalline silicon solar cells. Compared to the commercialized homojunction silicon solar cells, SHJ solar cells have higher power conversion efficiency, lower temperature coefficient, and lower manufacturing temperatures. Recently, several new record efficiencies have been achieved. To meet the continued demand for high-efficiency solar cells, expectations for large-scale mass production of SHJ solar cells are rising. To approach the efficiency limit and industrialization of SHJ solar cells, serious attempts have been made, yielding higher short-circuit current, open-circuit voltage, and fill factor. In this article, these recent advancements are reviewed, which reveals the future roadmap for approaching the efficiency limit.

1. Overview Introduction

1.1. Development Status of Silicon-Based Solar Cells

Global climate change has already had observable effects on the environment. The development of renewable energy technologies needs to be accelerated to reduce the consumption of fossil energy and realize carbon neutrality. One of the most optimistic renewable energy sources is photovoltaic (PV) solar energy, which is currently dominated by various crystalline silicon PV technologies, according to data of ITRPV 2020 results (≈95% of 135 GW total in 2020). Because of its advantages such as high power conversion efficiency (PCE), long stability, non-toxic and abundant materials, as well as its well-developed, scalable mass production techniques, silicon-based PV solar energy could represent one of the main power generation sources by 2050.

The maximum PCE of silicon solar cells, which is limited by the intrinsic properties of silicon, is of interest for PV researchers. In 1961, W. Shockley and H. J. Queisser calculated the bandgap dependent efficiency limit of solar cells using the detailed balance method in this case, and only hole–electron pairs' radiative recombination mechanisms were considered.[1] The maximum efficiency was found to be 30% for crystalline Si-based solar cells with an energy gap of 1.1 eV. In 1983, Tiedje et al. recalculated the efficiency limit of 29.8% for silicon solar cells based on the measured optical absorption spectrum and published values of the Auger and free carrier absorption coefficients.[2] The widely accepted efficiency limits for crystalline silicon solar cells with Lambertian light trapping under 1-sun was calculated to be 29.43% for a 110-µm-thick device, by using improved data sources (e.g., the light spectrum) and the commonly applied weak absorption approximation for light trapping,[3] or 29.56% for a 98.1-µm-thick device while using the ideal Lambertian light trapping.[4] Schmidt et al. compared the capability of surface passivation and carrier selectivity of the various options of carrier-selective layers (e.g., high doping layers, amorphous silicon layers, poly-Si layer, etc.) and analyzed their efficiency potentials.[5] For solar cells with amorphous silicon carrier-selective passivation contact layers, maximum efficiency of 27.5% could be reached. Very recently, the theoretical efficiency limit of solar cells with this structure was estimated to be 28.5% by Long et al. with the updated contact resistivities.[6] These theoretical values show how close Si-based solar cells are to the efficiency limit. Therefore, further improvements must be made to the manufacturing process of Si-based solar cells to get closer to the efficiency limit.

The parameters of a c-Si solar cell concept involve three factors. First of all, the number of photons absorbed by the c-Si solar cell must be maximized to generate as many carriers as possible. Recently, a high short-circuit current density J_{SC} (42.9 mA cm^{-2}) has been achieved in front-junction tunnel oxide passivating contacts solar cells (n-type TOPCon) by using c-Si wafers with a thickness of 200 µm.[7] Second, the concentration of electrons and holes held in the solar cell must be maximized, as their concentration under light induces the splitting of the quasi-Fermi level which determines the cell's open-circuit

Z. Sun, Y. He, J. Li, J. Wang, H. Yan
Faculty of Materials and Manufacturing
Beijing University of Technology
Beijing 100124, China
X. Chen, Y. Zhang
Faculty of Materials and Manufacturing
Key Laboratory of Optoelectronics Technology of Education Ministry of China, Faculty of Information Technology
Beijing University of Technology
Beijing 100124, China
E-mail: yzzhang@bjut.edu.cn

The ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/aenm.202200015.

DOI: 10.1002/aenm.202200015

voltage V_{OC}. Its theoretical value of 760 mV can be realized by minimizing the extrinsic recombination loss through perfect surface passivation. The recombination loss is quantified by the recombination current density parameter J_0 (femtoampere per centimeter square).[8] However, the J_{SC} and the V_{OC} of Si solar cells depend strongly on the wafer thickness. The limit of J_{SC} can be higher than 43.3 mA cm^{-2} if the wafer becomes thicker than 100 µm, and the limit of V_{OC} can be higher than 760 mV if the wafer becomes thinner. Therefore, there exists an optimum wafer thickness to achieve the highest efficiency. Finally, to generate an electrical current, it is necessary to separate the electrons and the holes and extract them along different paths. This is achieved by the emitter and the back surface field (BSF) of solar cells, in which minority carriers and majority carriers are respectively extracted. The performance of the emitter and BSF is characterized by its contact resistance ρ_C and its selectivity S_{10}, which influence the fill factor parameter FF of solar cells.[5]

Among silicon solar cells technologies, the aluminum back surface field (Al-BSF) solar cells and passivated emitter and rear cells (PERC) predominate in large-scale industrial PV device production. In the Al-BSF and PERC solar cells, the photogenerated minority carriers are collected by emitters formed by a dopant diffusion process. The main difference between these two structures is at the rear side where the full area aluminum alloyed back contact is replaced by a dielectric passivation layer, such as an aluminum oxide layer, and local aluminum alloyed contacts (as shown in **Figure 1**a,b). This evolution decreases the recombination current density of the rear side. However, due to the use of silicon-metal contacts on the front and rear sides, severe recombination current density (400–500 fA cm^{-2} at the metal contact area) results in a loss in V_{OC} (as data shown in **Figure 2**). Many new strategies have been proposed to address this issue. For example, a promising route on PERC is the TOPCon solar cells, which were introduced by Feldmann et al. at Fraunhofer ISE in 2013.[9] The main difference between PERC and TOPCon is at the rear where the local metal contact is replaced by layer stack of SiO_x/poly-Si(n^+), resulting in a smaller recombination current density (6 fA cm^{-2}) than that of PERC (36 fA cm^{-2}) as data shown in Figure 2. The advantage of this structure is that electrons are efficiently extracted without compromising the surface passivation and ohmic contact formation. This concept is named selective passivating contact.[10]

Besides these homojunction solar cells, another promising route is silicon heterojunction solar cells (SHJ), as shown in Figure 1d. This Heterojunction with Intrinsic Thin layer technology (HIT) was invented by Sanyo in the late 1980s. The backward current density of dark $I–V$ characteristics was significantly reduced from ≈10^{-6} to ≈10^{-8} A cm^{-2} with this HIT structure.[11] In Figure 2, we can see that the most remarkable advantage of amorphous silicon/c-Si heterojunction is the lowest recombination current density value among these contacts. Sanyo had improved the optical and electrical properties of the transparent conductive oxide (TCO) layer, the metal grid electrode, and the amorphous silicon layers, resulting in a continuously increased efficiency. In the last two decades, this technology is attracting an increasing amount of attention worldwide and its conversion efficiency is growing rapidly. As shown in **Figure 3**, the conversion efficiency of this kind of solar cell has grown from 20.1% up to 26.3% in the last two decades. In 2015, KANEKA created a world record (25.1%) of both-side-contacted c-Si solar cells.[12] In 2019, the efficiency was rewritten by a Chinese enterprise Hanergy, benefitting from rear emitter structures and optimized carrier transport layers.[13] Very recently, several breakthroughs have been achieved based on this technique. In September 2021, a record efficiency (25.54%) with a J_{SC} of more than 40 mA cm^{-2} has been reported by the equipment manufacturer Maxwell and Cu-plating technique start-up company Sundrive. One month later, LONGi pushed the conversion efficiency of SHJ solar cells to over 26% (Figure 3).[14] The high performance of SHJ solar cells comes from the superior surface passivation with thin intrinsic hydrogenated amorphous silicon layers (a-Si:H(i)) on both front and rear sides, rather than only the rear side. As a result, V_{OC} over 750 mV and FF over 85% are achieved using this type of structure.

Since 2014, the efficiency records for crystalline silicon-based solar cells under 1-sun illumination belong to heterojunction interdigitated back-contact solar cells (HJ-IBC).[15] In 2017, KANEKA realized the highest efficiency (26.7%) in c-Si solar cells by using this concept.[16,17] The highest efficiencies for silicon solar cell technologies (Al-BSF, PERC, TOPCon, SHJ, and heterojunction interdigitated back-contact solar cells [HJ-IBC])

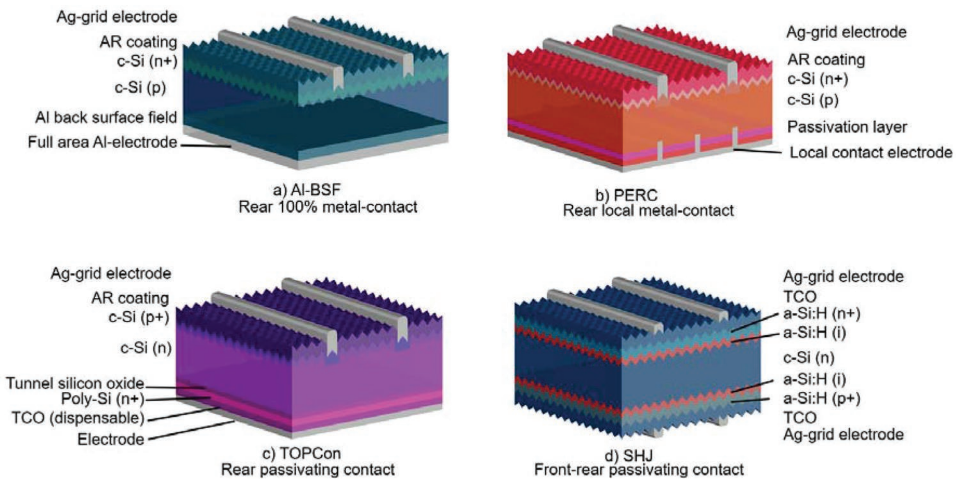

Figure 1. Schematic structures of the four dominating solar cells. a) Al-BSF; b) PERC; c) n-type TOPCon; d) SHJ.

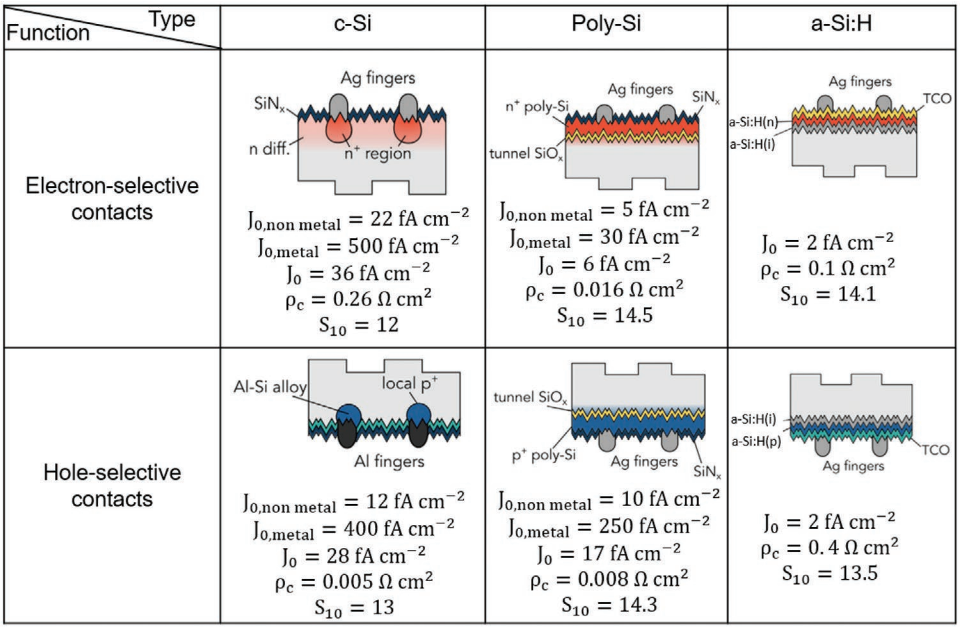

Figure 2. Comparison of junction formation technologies based on dopant diffusion (c-Si), polycrystalline silicon (Poly-Si), or amorphous silicon (a-Si:H), along with their parameters including the recombination current density J_0, contact resistance ρ_C, and carrier selectivity S_{10}. The corresponding data of contact resistance and carrier selectivity are obtained from references.[5] The inset figures are reproduced with permission.[25] Copyright 2021, Elsevier.

are compared in **Table 1**. Compared with the theoretical efficiency limit, there is still a noticeable gap for the further development of SHJ solar cells in the coming years. Additionally, the manufacturing cost of SHJ solar cells is comparatively higher than PERC solar cells because of higher raw materials costs (silver paste, indium, n-type silicon wafer). The levelized cost of electricity (LCOE) is defined as the net present value of the unit cost of electricity over the lifetime of the PV system and is a crucial parameter for industrialization:

$$\mathrm{LCOE}(\eta, C_m) = \frac{I + \sum_{t=0}^{N} \frac{OM}{(1+r)^t}}{\sum_{t=0}^{N} \frac{EY(1-deg)^t}{(1+r)^t}} \tag{1}$$

where I is the initial investment for installation of the system, OM are annual operation and maintenance expenses, EY is the energy yield in the first year of the system in kilowatts per hour, N is the total project lifetime in years, deg the annual module degradation rate, and r the nominal discount rate for the customer.[18] Another research focus for this advanced technology is to decrease manufacturing costs to decrease the initial investment. As a result, to obtain competitiveness in the LOCE, the PCE should be raised and at the same time manufacturing costs should be reduced. In this review, solutions to achieve higher conversion efficiency as well as to decrease manufacturing costs will be discussed.

1.2. Specification of SHJ Solar Cells

The most prominent features of SHJ solar cells are very high efficiency (>26%) and relatively simple fabrication processes.[14] Good temperature coefficients (approximately −0.2 to −0.27% °C^{-1}) are an appealing point of SHJ solar cells, which is essential in hot and sunny climates.[19,20] Additionally, SHJ solar cells manifest high bifaciality (>95%), which yield more electrical power generation.[21] SHJ solar cells consist of a symmetrical structure as demonstrated in Figure 1d. It is composed of a c-Si wafer, stacks of thin layers including silicon-based layers, TCO layers, and metallic grids.

Usually, Si(100) substrates are used for SHJ solar cell fabrication. The wafers are first cleaned and etched to remove saw damage. Subsequently, the wafers are textured with an alkaline solution: due to bond-density-dependent crystal dissolution, Si(111) faceted pyramids are revealed.[22] After texturization, an RCA cleaning process is applied to ensure a clean surface. Following cleaning and texturing preparation, intrinsic amorphous silicon layers (a-Si:H(i)) and doped amorphous silicon layers (a-Si:H(p) and a-Si:H(n)) are deposited by chemical vapor deposition (CVD). These silicon-based layers are usually deposited by plasma- enhanced CVD process (PECVD) at low temperature (<200 °C).[23] Another technique that has been used for SHJ production is catalytic chemical vapor deposition (cat-CVD).[24] TCO layers on the front and rear sides are deposited by a physical vapor deposition process (PVD), either magnetron sputtering or reactive plasma deposition. Finally, the electrodes are made by methods such as screen printing, where a low-temperature curing silver paste is usually printed.

2. Work Principle of SHJ Solar Cells

In solar cells, solar energy is converted to electric energy. Photons must transmit the TCO layers and a-Si:H layers, and then are absorbed by the silicon wafer in SHJ solar cells. The

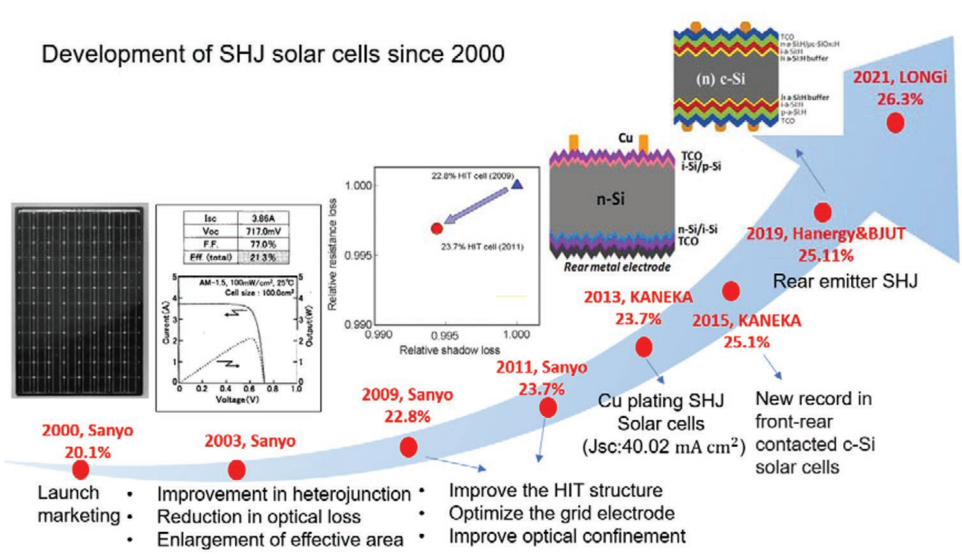

Figure 3. Efficiency achieved over the last two decades for SHJ solar cells by Sanyo. Reproduced with permission.[26] Copyright 2000, John Wiley and Sons; Reproduced with permission.[27] Copyright 2003, IEEE; Reproduced with permission.[29] Copyright 2011, EUPVSEC. KANEKA, Reproduced with permission.[12,30] Copyright 2015, AIP Publishing. Hanergy, Reproduced with permission.[13] Copyright 2020, Elsevier and LONGi.[14]

photogenerated electrons and holes are extracted by carrier selecting transportation layers. The properties of each part (passivation layers, silicon bulk, and carrier selective transportation layers) and their effects on the performance of SHJ solar cells will be reviewed in the following section.

2.1. Passivation Layers: Passivation by a-Si:H(i)

After texturization treatment of a (100) surface silicon wafer, one atom at the surface contributes a one half-saturated dangling bond (DB), sticking out perpendicular from the surface and bonded to three Si atoms (as shown in **Figure 4**). The surface states induced by DB are half-filled and lie at mid-gap.[34] The DB electronic structure at the surface gives rise to surface states within the bandgap of c-Si. In c-Si solar cells, the recombination centers at the surface area originate from DB localized in a very small interlayer extending over only a few angstroms.[35] To solve the DB problem, dielectric passivation layers cover the majority of the front and rear surfaces for PERC solar cells. Dielectric layers on the front surface and a thin silicon oxide layer on the rear surface are applied to TOPCon solar cells. In contrast, in SHJ solar cells, an intrinsic hydrogenated amorphous silicon layer is embedded between the c-Si and the doped a-Si:H on both sides. After passivation by a-Si:H, the density of DB can be reduced from 10^{15} cm^{-2} down to 10^{8} cm^{-2}.[36,37] This results in a long effective carrier lifetime τ_{eff}. As the total recombination loss includes bulk recombination and surface recombination, the τ_{eff} is described as Equation (2):

$$\frac{1}{\tau_{\mathrm{eff}}} = \frac{1}{\tau_{\mathrm{bulk}}} + \frac{1}{\tau_{\mathrm{surf}}} \tag{2}$$

where the intrinsic a-Si:H layers improve the τ_{surf}. Apart from τ_{eff}, another way to evaluate the recombination loss is analyzing the recombination current density J_0 (fA cm^{-2}). According to recombination mechanisms, the total recombination current density is described as Equation (3):

$$J_0 = J_{0,\mathrm{radiative}} + J_{0,\mathrm{Auger}} + J_{0,\mathrm{SRH}} + J_{0,\mathrm{surf}} \tag{3}$$

Therein the surface recombination current density $J_{0,\mathrm{surf}}$ is expressed as Equation (4):[38]

$$J_{0,\mathrm{surf}} = \frac{q n_i^2 S_{\mathrm{eff}}}{N_{\mathrm{D}}} \tag{4}$$

where $J_{0,\mathrm{surf}}$ is determined by the surface recombination velocity S_{eff}, and doping concentration N_{D}. The recombination parameters (τ_{eff} and J_0) can be obtained by photoconductance decay measurements. For SHJ solar cells, the values of τ_{eff} and $J_{0,\mathrm{surf}}$ can be several milliseconds and less than 2 fA cm^{-2}.[39]

Table 1. Summarized selected certified conversion efficiencies for Al-BSF, PERC, TOPCon, SHJ, and HJ-IBC.

Solar cell type	Manufacturer	Efficiency	Area [cm^2]	Reference
Al-BSF	Shinsung Solar Energy	20.29%	239.1	[31]
PERC	UNSW	25%	4	[32]
n-type TOPCon	FhG ISE	25.8%	4	[7]
p-type TOPCon	FhG ISE	26%	4	[33]
SHJ	LONGi	26.3%	274.5	[14]
HJ-IBC	KANEKA	26.7%	180	[16]

2.2. Quality of Monocrystalline Silicon Wafer

Up to now, the highest efficiency SHJ solar cells were obtained using n-doped, rather than p-doped silicon wafers, mainly due to the lower minority carrier lifetime for p-doped

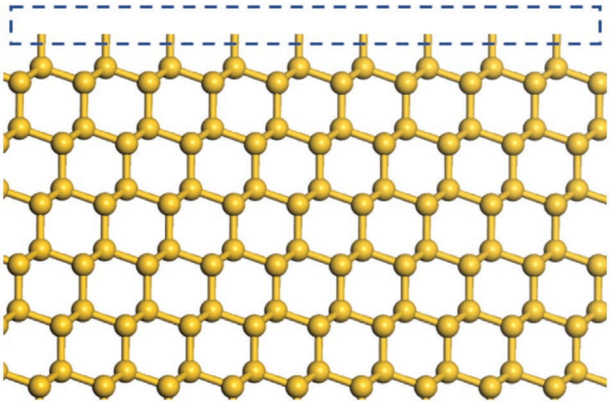

Figure 4. Dangling bonds (circled by the blue rectangle) of (111) orientation silicon surface. These dangling bonds must be passivated to reduce the carrier recombination.

silicon wafers.[40] Parameters such as wafer thickness, doping concentration, and impurity density influence the performance of solar cells. Its thickness not only influences the absorption of photons with long wavelengths (>1000 nm) but also the transportation of carriers.[41] In the next section, the effects of impurities will be discussed.

By improving the conversion efficiency through optimized structure and fabrication processes, the final device efficiency becomes increasingly sensitive to the contaminants in the silicon wafer.[42] The minimization of recombination or maximization of carrier lifetime in the bulk (τ_{bulk}) is a key factor in obtaining high-efficiency solar cells. The bulk recombination mechanisms include Auger recombination, radiative recombination, and Shockley–Read–Hall (SRH) recombination, and therefore could be described as Equation (5):

$$\frac{1}{\tau_{\text{bulk}}} = \frac{1}{\tau_{\text{Auger}}} + \frac{1}{\tau_{\text{radiative}}} + \frac{1}{\tau_{\text{SRH}}} \tag{5}$$

Among these recombination mechanisms, Auger recombination and radiative recombination are considered as intrinsic properties, while the SRH recombination determined by the bulk defects is critical for device performance. As the τ and J_0 are interchangeable with each other, the Equation (5) can be written as Equation (6):

$$J_{0,\text{bulk}} = J_{0,\text{Auger}} + J_{0,\text{radiative}} + J_{0,\text{SRH}} \tag{6}$$

Therein for an n-type wafer, its recombination current density caused by SRH is expressed as Equation (7):[8]

$$J_{0,\text{SRH}} = q\frac{n_0 W}{\tau_{\text{eff}}} = q\frac{n_i^2 W}{N_{\text{D}}\tau_{\text{minority}}} \tag{7}$$

where n_0, n_i, N_{D}, and W denote the minority carrier concentration, the intrinsic carrier concentration, the doping concentration and the wafer thickness respectively. Hence, improved wafers (longer carrier lifetime τ_{eff} or τ_{minority}) are essential to achieve higher efficiency. In Equations (4) and (7), J_0 signifies the recombination current in equilibrium, which usually varies with the excess carrier concentration.

Due to cost constraints, silicon materials for PV inherently contain a significant amount of metal impurities and defects, further contamination can occur during the fabrication process. In the usually used n-type c-Si silicon wafers for SHJ solar cells, phosphorus atoms are intentionally added as donors. Impurities such as oxygen and metal atoms originate from the raw materials, processing, or crystal growth.[43] Chemically, the majority of the metal atoms are in the form of metal-silicide precipitates or metal-rich clusters, which are often attached to extended defects.[42,44] In conventional mc-Si materials, the total (precipitated and dissolved) concentrations of Fe, Cu, Ni, Cr, and Co were reported to be generally on the order of 10^{13} and 10^{14} cm^{-3}.[45] Electrically, most of the metal atoms form deep energy levels in the bandgap of silicon and therefore enhance the recombination of electron–hole pairs.[46] The defects induced by oxygen for n-type silicon solar cells are different from conventional p-type silicon solar cells. Specifically, instead of forming a boron-oxygen defect with the dopant under illumination or current injection for p-type silicon solar cells, the defects associated with oxygen in n-type silicon are thermal donors (TD).[47,48] A deep energy level resulting from thermal donors is in the range of 0.2–0.3 eV below the conduction band, and the τ_{eff} extracted at $\Delta n = 5 \times 10^{14}$ cm^{-3} is decreased from ≈3 to ≈1.5 ms when the concentration of TD is 10^{15} cm^{-3}.[48] The effect of concentration of oxygen on the conversion efficiency of SHJ solar cells was studied by Meng.[49] A lower oxygen concentration was proved to be necessary to obtain high *FF* and high efficiency of SHJ solar cells. As the distribution of impurities in an ingot isn't homogeneous, Zhao et al. investigated the stability of SHJ solar cells in mass production relating to process variations and incoming wafer quality.[50] The metal impurities often appear in tail parts of ingots and the oxygen atoms mostly observed in seed parts of ingots can influence the bulk lifetime.

2.3. Carrier Selective Layers: Selection and Extraction of Carriers

Photogenerated electrons and holes should be efficiently and selectively collected by the corresponding electrodes to generate electronic current. In SHJ solar cells, asymmetrical heavily doped a-Si:H layers are used to realize this selective collection. As the bandgap of amorphous silicon (≈1.7 eV) is larger than crystal silicon (1.1 eV), band offsets are present at the a-Si:H/c-Si interface, which is ≈0.44 eV for the valence band and ≈0.15 eV for the conduction band. As can be seen in **Figure 5**, at the contact region, the band bending leads to the repelling of the electrons at the emitters and the holes at the BSF. This contact structure consisting of an intrinsic passivation layer and a heavily doped overlayer that separates the metal contacts from the silicon wafer is named passivated contact. This passivated contact structure simultaneously fulfills the passivation and contacting roles, providing a high level of surface passivation as well as contrasting conductivities for the electrons and holes.[51–53]

Besides the passivating layers and the carrier selective transport layers in SHJs, TCO layers are needed to provide lateral conductivity. The TCO is crucial in amorphous/crystalline silicon heterojunction solar cell performance as they have to be highly conductive to ensure the transport of photogenerated charges as well as being highly transparent for the sun's

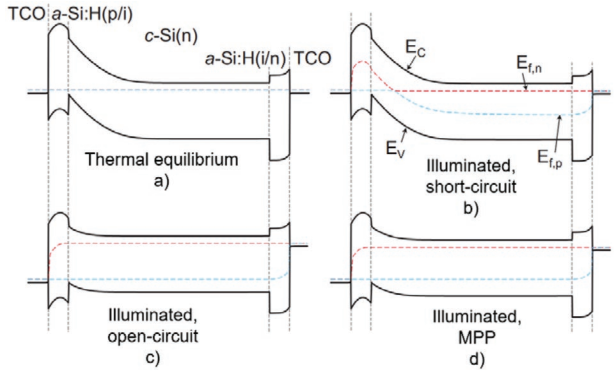

Figure 5. Energy band diagrams of a SHJ device. a) In thermal equilibrium; b) illuminated short-circuit state; c) illuminated open-circuit state; and d) illuminated maximum power point (MPP). With the conduction band edge E_C, valance band edge E_V, and the quasi-Fermi-levels for electrons $E_{f,n}$ and holes $E_{f,p}$, respectively. Reproduced with permission.[59] Copyright 2018, Elsevier.

spectrum.[54] The TCO layer can mainly be divided into 2 types: indium oxide-based TCO and zinc oxide-based TCO. Indium tin oxide (ITO, In_2O_3:Sn) is a widely used TCO material for SHJ solar cells.[54–56] As the lateral transportation is promoted by the TCO, the enlarged spaces of finger electrodes result in a high bifaciality (>95%).[21,26,57] Its main limitation for applications in SHJ is parasitic absorption in the IR range caused by free carriers.[54,58] Its electrical parameters such as carrier mobility, carrier concentration, and work function, as well as the optical parameters such as refractive index and thickness, influence the performance of solar cells.

3. Roadmap toward Efficiency Limit of SHJ Solar Cells

As introduced above, the efficiency limit of silicon solar cells is 29.4% with an open-circuit voltage of 761 mV, short-circuit current density of 43.3 mA cm^{-2} and fill factor of 89.3%.[3] The efficiency loss results from 3 main loss mechanisms: optical loss, recombination loss, and transport loss. As shown in **Figure 6**, power loss analysis (PLA) relative to the theoretical efficiency limit is helpful for a deeper understanding of these loss mechanisms.[17,33] First, the optical PLA can be evaluated directly from the measured EQE and reflectivity. Regarding SHJ solar cells, optical losses including reflectance, transmittance, parasitic absorption, and shading losses, decrease the short-circuit current.[60] Second, the PLA of transportation can be determined by *FF* analysis. Since the pseudo-*FF* parameter obtained from Suns-V_{oc} measurement is unaffected by the voltage drop due to series resistance,[61] the portion of loss caused by series resistance can be obtained by subtracting the light *J–V* curve *FF* from the pseudo-*FF*. The difference between this pseudo-*FF* and theoretical *FF* limit (89.3%) is attributed to carrier recombination. Third, the free energy loss analysis (FELA) introduced by Brendel et al. can be used to obtain the electrical loss including recombination and transportation.[62,63] The starting point of free energy loss analysis (FELA) is that electrical power generated by a solar cell device is free of entropy. The transport losses and recombination losses can be expressed as a rate of free energy (milliwatts per centimeter square). These quantities can be expressed directly in absolute efficiency units (percentage) by normalizing them by the total power density of the AM1.5G spectrum. The recombination losses in the bulk can be calculated by Equation (8):

$$\varnothing_{\text{bulk}} = A^{-1} \iiint R\left(E_{f,n} - E_{f,p}\right) dV \tag{8}$$

in which R is the total recombination rate including Auger, SRH, and radiative recombination in bulk, $E_{f,n}$ and $E_{f,p}$ are the quasi-Fermi energy levels for electrons and holes (unit in volts), dV represents the volume integration, and A being the surface area.

The recombination losses at each contact interface can be expressed by Equation (9):

$$\varnothing_{\text{surface}} = A^{-1} \iint J_0\left(E_{f,n} - E_{f,p}\right) dA \tag{9}$$

in which J_0 is the recombination current density at the surface and dA represents the surface integration. Additionally, the resistive loss for electrons and holes is expressed by Equation (10):

$$\phi_{\text{transport}} = A^{-1} \iiint J^2/\sigma dV \tag{10}$$

in which J represents the current density of carriers and σ represents the conductivity.

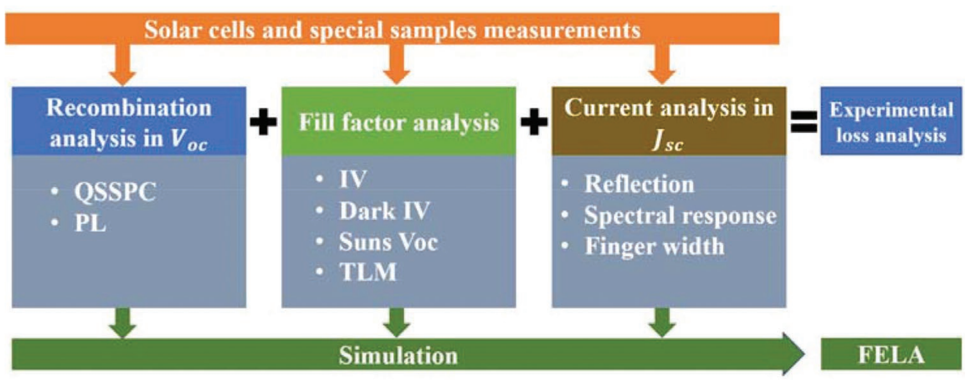

Figure 6. Schematic representing the approach for the loss analysis. Reproduced with permission.[64] Copyright 2017, Wiley-VCH.

In the following part, roadmaps toward higher efficiency will be discussed in 3 subsections: short-circuit current J_{SC}, open-circuit voltage V_{OC}, and fill factor *FF*. As the J_{SC} is a weakness for SHJ solar cells, first roadmaps toward higher J_{SC} will be discussed, followed by J_{SC} roadmaps toward higher V_{OC}, and finally roadmaps toward higher *FF*.

3.1. Roadmap toward Higher J_{SC}

A large short-circuit current requires a solar cell to absorb and convert sunlight into electrical energy at optimal levels. However, there are reflection losses at the front surface, transmittance loss at the rear surface, and absorption loss such as parasitic absorption by amorphous silicon and free-carrier absorption by the TCO layers. Additionally, the metal electrode at the front side shades a proportion of sunlight. In this next section, solutions to decrease these optical losses will be discussed.

3.1.1. Reduction of Parasitic Absorption in Amorphous Silicon

The absorption of sunlight by the hydrogenated amorphous silicon does not efficiently contribute to J_{SC} and therefore results in so-called parasitic absorption. In SHJ solar cells, Holman et al. estimated that light absorbed in doped a-Si:H layers does not contribute to photocurrent, but about one third of the light absorbed in the a-Si:H(i) layer does contribute to the photocurrent.[65] Similarly, carrier injection efficiencies of 40% to 42% were found for a-Si:H(i) layers capped with a-Si:H(p) layers and the carrier injection was found to be ≈100% from a-Si:H(i) into c-Si for samples without additional capping doped layers.[66]

A method to suppress parasitic absorption is to use doped hydrogenated nanocrystalline silicon (nc-Si:H) instead of doped a-Si:H. The reason for nc-Si:H being the material of choice and being preferred over a-Si:H for SHJ application can be explained based on the difference in their absorption coefficient. The crystallographic structure plays a significant role. The well-defined crystalline structure means well-defined bond lengths and bond angles in nc-Si:H (long-range disorder, short-range order), representing well-defined band-to-band absorption. On the contrary, disturbing bond lengths and bond angles in a-Si:H (short-range disorder) contribute to the internal scattering of light. Therefore, nc-Si:H layers can suppress parasitic absorption in the visible light range.[67] When replacing the doped a-Si:H layers by nc-Si:H contact layers, two aspects have to be considered: the effect of their deposition on the a-Si:H(i) passivation layer, and the necessity to reach a high crystalline fraction.[68] The initial stages of growth of nc-Si:H layers are the most critical because they influence the crystalline fraction in the layer. The epitaxial regions of the a-Si:H(i) layer behaves as a nucleation layer for the nanocrystalline growth. For solar cells with eip.-free a-Si:H(i) films, Mazzarella et al. proved that a CO_2 plasma treatment before nc-Si:H emitter deposition could yield a high voltage and a high fill factor.[68] In 2015, Seif et al. has reviewed and discussed different strategies to facilitate the nucleation of nanocrystalline silicon layers and assess their compatibility with SHJ solar cell fabrication, including CO_2 treatment, inserting buffer layers, and gas and temperature variations.[69]

Alloying with oxygen, carbon, or nitrogen help to further reduce parasitic absorption. Another route is to employ doped wide-bandgap silicon alloys such as a-SiC$_x$:H,[70] a-SiO$_x$:H,[71] or nc-SiC$_x$:H.[72] Wider bandgap silicon-oxygen alloy (nc-SiO$_x$:H)[71,73–78] has been studied and has shown great potential to replace a-Si:H, given its low optical absorption coefficient and tunable refractive index. The SiO$_x$:H layer can be deposited by the PECVD process with silane (SiH_4), hydrogen (H_2), phosphine (PH_3), and carbon dioxide (CO_2) as precursors, in which CO_2 acts as the oxygen source. This SiO$_x$:H layer could be amorphous (a-SiO$_x$:H) or nanocrystalline (nc-SiO$_x$:H). Zhang et al. have used n-type a-SiO$_x$:H layer to replace doped a-Si:H layers.[71] They found that the doping efficiency of phosphorus in the a-SiO$_x$:H layer was lower than that in a-Si:H, which was detrimental to the extraction of photogenerated electrons and the field-effect passivation.[71] Doped hydrogenated nanocrystalline silicon oxide films (nc-SiO$_x$:H) have been suggested to overcome these limitations. However, the deposition of high-quality nc-SiO$_x$:H layer is a bottleneck because its adjacent intrinsic a-Si:H layer inhibits the nucleation and crystallization of the nc-SiO$_x$:H layer.[68,77] To improve the crystallinity, a seed layer for fast nucleation is necessary. Addressing this problem, Peng et al. have demonstrated that high phosphorus-doped seed layers (with 10% PH_3 gas during deposition) can promote nucleation and crystallinity.[79]

While using wide bandgap silicon alloys as carrier contact layers, unbeneficial alignment of energy bands could induce carrier transport loss by increasing contact resistance. These substituted layers could be either used alone as carrier selective contacts or combined to minimize the transport loss.[80] This issue will be discussed in the following section associated with the fill factor.

3.1.2. Reduction of Parasitic Absorption in TCO

Unlike the parasitic absorption caused by amorphous silicon in the visible spectrum, the spectral loss caused by TCO is free-carrier absorption in the range of near-infrared. The key parameter to improve the transparency/conductivity of TCO is carrier mobility since it allows increasing conductivity without losing transparency in the near-infrared region.[54,56]

For indium oxide-based TCO layers, various cation-doped In_2O_3 have been investigated so far in SHJ solar cells. To date, tin-doped and tungsten-doped In_2O_3 films have been applied in industrial SHJ solar cells, though probably they also contain a significant amount of hydrogen as a second dopant. ITO layers grown at temperatures below 200 °C, which reach electron mobility (μ) ≈ 40 cm^2 V^{-1} s^{-1} and carrier concentration ≈1.5 × 10^{20} cm^3, are the most widely applied TCO materials.[81] ITO layers are usually deposited via magnetron sputtering. The ratio of indium:tin plays an important role in its optical performance. The ITO with lower tin content (97:3) shows a better short-wavelength optical advantage.[82] It is known that oxygen vacancies contribute to the free carrier density in TCOs, so the oxygen partial pressure is a key factor for optical performance during deposition.[83] In_2O_3 based TCOs doped with tungsten (IWO) show high values of μ greater than 80 $cm^2\ V^{-1}\ s^{-1}$ at a charge carrier density ranging from 1×10^{20} to 3×10^{20} cm^{-3}.[84] Benefiting from its high electron mobility, IWO layers not only yield a low resistivity but also show high NIR transmittance (94.21% at 80 nm thickness).[85] Even higher mobility (μ > 100 $cm^2\ V^{-1}\ s^{-1}$) of TCO layers can be achieved by solid-phase crystallized (SPC) hydrogen-doped In_2O_3 (I_2O_3:H).[86,87] This was attributed to the appearance of hydrogen atoms as single charged donors, and their ability to passivate defects at the grain boundaries of polycrystalline films, as well as their beneficial role in the crystallization process of solid-phase crystallized (SPC) TCOs. By SPC processes, I_2O_3:H layers are first deposited at low temperatures (room temperature) in an amorphous matrix and subsequently annealed at temperatures ≈170 °C. Recently, high-mobility and thin NIR-transparent In_2O_3 films have been produced by doping metals such as Cerium,[88,89] Titanium,[90] and Zirconium,[91] which perform as transparent electrodes for solar cells having sensitivity in the visible to the NIR wavelength region.[92] With higher carrier mobility (>100 $cm^2\ V^{-1}\ s^{-1}$), In_2O_3:Ce, In_2O_3:Ti, and In_2O_3:Zr are promising TCO candidates for SHJ solar cell production.

In today's SHJ solar cells, the p-n junction is usually placed at the rear side of the wafer. Placing the junction on the rear side allows the wafer to support the front side TCO to laterally transport the majority carriers and, in turn, lessens the requirements on the conductivity of the front TCO.[93–95] To benefit from this lateral carrier transport, the structure of SHJ solar cells can be modified to obtain a high short-circuit current density. In 2021, Li et al. developed a TCO-free SHJ solar cell to decrease the free-carrier absorption loss caused by the TCO and lower costs.[96] This modification yielded a short-circuit current density of more than 40 mA cm^{-2}. In 2019, a number of groups demonstrated that SiN_x/TCO composite layers at the front surface for rear-emitter SHJ solar cells can enhance the optical performance without damaging the lateral transport of carriers and the passivation.[93,97,98] In this architecture the parasitic absorption of the TCO can be reduced by designing a thinner TCO layer than the single-layer anti-reflective optimum (≈80 nm). In this case, a second anti-reflecting coating such as SiN has to be deposited on top of the TCO to minimize reflection losses.

3.1.3. Reduction of Grid Shadowing

Reducing the metal grid shadowing is also necessary to improve the J_{SC}. Electrodes with high conductivity allow for decreasing the width of fingers and busbars. As high temperature above 200 °C damages the passivation of a-Si:H layers, the high temperature curing silver paste used on homojunction silicon solar cells is no more compatible with SHJ solar cells. Therefore, low-temperature curing silver paste is commonly used for SHJ solar cells. The challenge with low-temperature pastes is to achieve high conductivity while maintaining low contact resistance to the underlying TCO. Generally, a larger portion of low-temperature silver paste for better Ohmic contact is required for SHJ solar cells than homojunction c-Si solar cells, for example, 200 mg per cell for SHJ while 150 mg for other silicon solar cells (158.75 × 158.75 mm^2 cell size). To solve this problem, alternative metallization routes have been developed such as multi-busbar,[99,100] smart wire,[101] interconnect-shingling,[102] and Cu-plating. Among these metallization routes, copper plating is regarded as an ideal alternative electrode solution and has been industrially proven.[30,103,104] The resistivity of copper-plated electrodes is ≈2–3 times lower than the printed silver electrode because of its void-free structure. In addition, the Cu plate electrode width can be lowered to 20 µm, much finer than screen-printed Ag grids (>30 µm), which significantly reduces the grid shading.[105] As early as 2015, Adachi et al. have reported a 25.1% efficiency top/rear contacted SHJ cell (152 cm^2 for aperture area) using a Cu electrode, for which the J_{SC} is 40.8 mA cm^{-2}.[12] The very recent recorded efficiency of 25.54% for SHJ solar cells reported by Maxwell and Sundrive demonstrates the significant advantage of Cu-plating.

3.1.4. New Architectures

Very recently, Long et al. have updated the selectivity of carrier transport layers, the contact resistivities, and the theoretical efficiency limit of SHJ solar cells, based on a very high-efficiency SHJ solar cell (25.11%).[5,6] An advantage of the usage of n-type nanocrystalline silicon-oxygen-alloy (nc-SiO_x:H(n)), the selectivity and the contact resistance are optimized. According to their evaluation, the contact fraction for electron-selectivity that maximizes the efficiency was only ≈24%, rather than 100%. To further reduce parasitic absorption loss caused by a-Si:H layers and TCO layers, a patterned structure is a potential solution (**Figure 7**a). In 2019, Huang et al. have designed a Heterojunction of Amorphous silicon and Crystalline silicon with Localized pn structure (HACL).[106,107] Instead of a structure where the emitter and back-field layer cover the entire surface, the area of the emitter region covers ≈3% of the area of the wafer in HACL. This new design yields a higher J_{SC} than conventional SHJ cells: 42.55 mA cm^{-2} for rear-emitter cells and 43.06 mA cm^{-2} for front-emitter cells. More interestingly, TOPCon electron selective contact (SiOx/poly-Si(n+)) shows better selectivity than SHJ's electron selective contact. For this, an optically-transparent front surface field (FSF) can be formed at the illuminated side either by field-effect passivation or by diffusion in hybrid SHJ solar cells (Figure 7b).[108] A patterned hybrid structure combining a-Si:H(i)/a-Si:H(p) with SiOx/poly-Si(n+) contacts can be greater in efficiency (28.9%).[6] How to realize this patterning design is a research focus in near future approaches for higher short-circuit current are summarized in **Table 2**.

3.2. Roadmap toward Higher V_{OC}

The V_{OC} is limited by the splitting of electron and hole quasi-Fermi levels under illumination, which is determined by the excess charge carrier density as shown in Equation (11):

$$V_{oc} = \frac{kT}{q} \ln\left[\frac{(N_D + \Delta n)\Delta n}{n_i^2}\right] \tag{11}$$

where the excess charge carrier density Δn is required to be maximized.

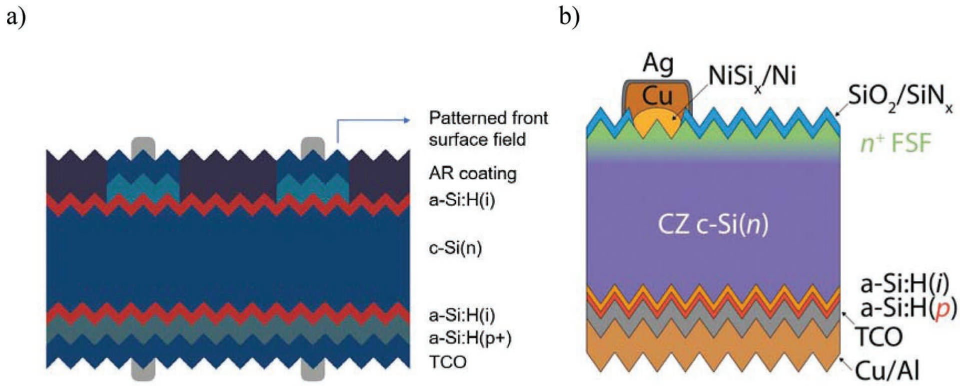

Figure 7. a) Schematic of SHJ solar cells with patterned front surface field (electron contact). This patterned structure at front side can reduce the parasitic absorption loss without affecting electron transport. According to Long's researches, the optimized contact fraction of a-Si:H(n) is 24.4%.[6] b) Schematic of n-type hybrid SHJ solar cells. Reproduced with permission.[19] Copyright 2017, Royal Society of Chemistry.

Losses in V_{OC} are mainly due to carrier recombination mechanisms (Equation 12):

$$V_{oc} = \frac{nkT}{q}\ln\left(\frac{I_L}{I_0}+1\right) \quad (12)$$

where the recombination current I_0 is required to be minimized. The greatest voltage can be achieved if there is only intrinsic recombination, such as radiative recombination and Auger recombination in the bulk. The extrinsic recombination such as defect-assisted recombination in the bulk and at interfaces should be suppressed. Two ways to yield higher excess charge carrier density are proposed: reducing the recombination rate of photogenerated carriers and decreasing the distance for excess carrier collection such as decreasing the thickness of silicon. The potential of thin silicon devices to reach high voltages (over 760 mV) in SHJ solar cells was demonstrated.[110] Sai et al. reported a detailed research on the wafer-thickness dependence of V_{OC}.[111] Thin crystalline silicon solar cells with a high V_{OC} and a high efficiency are very attractive for realizing lightweight and flexible wafer-based solar cells as well as for reducing material costs.[112]

Table 2. Summary of approaches for higher short-circuit current.

Approach for higher J_{SC}		
Optimization of a-Si:H	Wide bandgap carrier transport layers	a-SiC:H,[70] a-SiO$_X$:H,[71,73] nc-SiC:H,[72] nc-SiO$_X$:H[75–78]
	Indirect bandgap carrier transport layers	nc-Si:H[68,69]
Optimization of TCO	Higher mobility TCO layers	I$_2$O$_3$:W,[84,85] I$_2$O$_3$:H,[86,87] In$_2$O$_3$:Ce,[88,89] In$_2$O$_3$:Ti,[90] In$_2$O$_3$:Zr[91]
	TCO/SiO$_x$(SiN$_x$) composite layers	a-SiO$_2$/TCO,[93] SiO$_X$/I$_2$O$_3$:W,[98] SiN/ITO[97]
	TCO-free SHJ	Front TCO-free SHJ solar cells[96]
Reduction of grid shadowing	Cu-electroplated electrodes	Cu-grid electrodes[12,104,109]
	New silver finger design	Multi-busbar,[99,100] smart wire,[101] interconnect-shingling[102]
New architectures	Patterning a-Si:H layer	Local junction design[106]
	Hybrid solar cells	Hybrid SHJ solar cells[108]

3.2.1. Reduce Interfacial Recombination

Compared to other silicon solar cells (e.g., PERC, TOPCon), the silicon SHJ solar cells are superior in the passivation of c-Si surfaces, eliminating most of the recombination centers at surfaces. Mitigation of the remaining defect recombination losses is an obvious way to obtain higher V_{OC} values.[113] It is well known that an abrupt a-Si:H/c-Si interface at the c-Si surface is a key for good surface passivation.[114–117] In 2010, Descoeudres et al. studied the passivation effect by intrinsic a-Si:H layers deposited using pure silane. They demonstrated that the silane depletion fraction affected the minority lifetime, and good interface passivation was obtained from highly depleted silane plasma.[118] In 2011, Descoeudres et al. demonstrated a multiple-step method, namely, a pure silane plasma deposition followed by a hydrogen plasma annealing, which can improve the interface passivation.[119] A dual-intrinsic layer or bilayer approach is known to be effective to attain good surface passivation at the a-Si:H/c-Si interface.[120–122] Zhang et al. in 2017 and Ruan et al. in 2019 proposed the two-step method to obtain an epitaxial-free passivation bilayer (as shown in **Figure 8**).[13,123,124] Morales-Vilches et al. used a low hydrogen dilution plasma (H_2:SiH_4 = 1:1) to deposit the buffer layer at the initial state.[125] The buffer layer containing more SiH_2 molecules yields a higher minority lifetime. Very recently, Sai et al. studied various intrinsic bilayer deposition conditions, with a particular focus on the correlation between the growth condition and the material properties of the interfacial a-Si:H layer.[122,126] For a bilayer approach, the entire deposition process can be divided into the initial deposition stage and the growth stage. Yet, the success of the initial deposition state is critical to obtaining a sharp interface. At the initial stage, a very thin buffer layer was deposited by pure silane plasma while in the next stage, the silane was diluted by hydrogen to accomplish the deposition. The a-Si:H buffer layer could effectively suppress epitaxial growth as confirmed by HR-TEM. On top of the buffer layer is

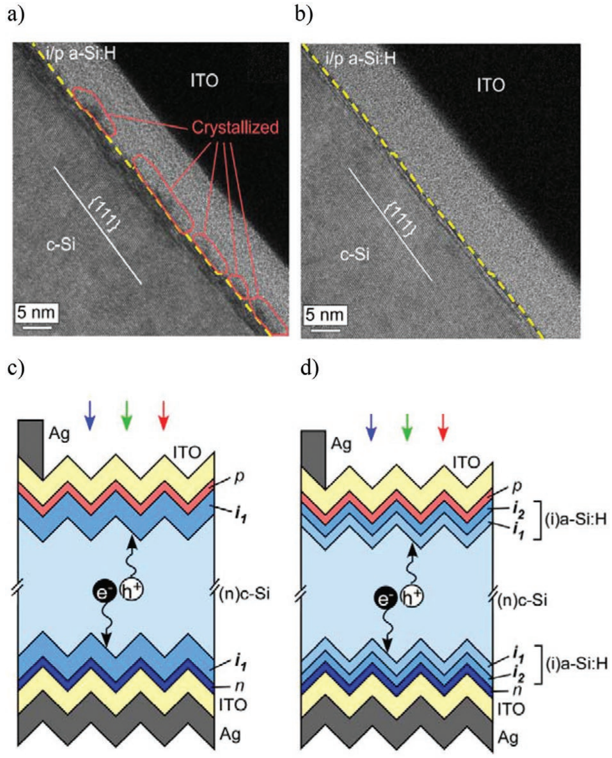

Figure 8. a) Cross-sectional TEM images near the p-side a-Si:H/c-Si interfaces of the SHJ solar cells with the single a-Si:H(i) layer. In this case, partial crystallization was observed near the interface, and b) the a-Si:H(i) bilayer ($i_1 + i_2$). An abrupt a-Si:H/c-Si interface was obtained by using the bilayer. Reproduced with permission.[126] Copyright 2018, AIP publishing. c) Schematic illustrations of front-junction SHJ solar cells prepared with single a-Si:H(i) layers, and d) two-step a-Si:H(i) layers or bilayers ($i_1 + i_2$). Reproduced with permission.[122] Copyright 2021, Wiley-VCH.

the main passivation layer, deposited by a high hydrogen dilution plasma. This main passivation layer is denser than the buffer layer.

Some researchers have proved that the plasma is detrimental to the passivation of the c-Si/a-Si:H interface because the plasma etches the c-Si surface. The plasma-free CVD deposition such as cat-CVD has been used to deposit high-quality passivation layers.[24,127] The thickness and sharpness of the epitaxial layer on the surface of c-Si can be determined by using conventional high-resolution TEM (HRTEM).[117,124,128] According to an investigation by Matsumura et al., the cat-CVD can realize very sharp a-Si:H/c-Si interfaces and thin transition-layers from c-Si to a-Si:H (<0.6 nm), which is 1.8 nm for PECVD.[127] Recently, more advanced TEM methods such as spherical aberration (CS)-corrected TEM techniques have revealed the atomic-level structure mechanisms of the epitaxial layers. Using the C_x-corrected TEM and atomic-resolution HAADF-STEM, Matsumura et al. revealed the morphology of transition layers from c-Si to a-Si:H with 0.08 nm spatial resolution.[127] Higashimine et al. analyzed the effect of the thicknesses of a-Si:H layers as well as the growth temperature on damaged formations with HAADF STEM.[24] Very recently, with the help of HAADF, Qu et al., for the first time identified nanotwins at the c-Si/a-Si:H interface in high-efficiency SHJ solar cells which could be categorized as either embedded nanotwins or free nanotwins.[129] The embedded nanotwins could deteriorate the device performance by inducing deep-level defects. In addition, they also found that these embedded nanotwins grow from the passivated interfaces in the subsequent high-temperature processes near 200 °C such as deposition of doped a-Si:H, TCO, or electrode sintering. This discovery highlighted that even in very high-efficiency SHJ solar cells, the interface quality still needs to be optimized.

3.2.2. Reduction of Bulk Recombination

As mentioned above, the impurities in silicon bulk should be minimized to obtain a sufficient minority carrier lifetime. Gettering is the process applied to remove metallic impurities to a less harmful region of the device.[130] Gettering can be considered as a three-step process: release of the defect, especially metallic impurities from its precipitate form, diffusion of defects to the gettering region, and capture of the defects at the gettering sites.[131] Higher process temperatures can dissolve metal precipitates and improve the diffusivity of the impurities toward the sink or capture sites near surfaces. The various capture approaches in the context of silicon PV include phosphorus diffusion, boron diffusion, as well as dielectric films deposition.

Phosphorus-diffusion gettering (PDG), a near-surface region of the wafer into which phosphorus is diffused from a deposited or gaseous source, affects a Fermi-level shift and metal-dopant pairing which are believed to be augmented by nonequilibrium defect-related processes.[132–134] For homojunction c-Si solar cells such as PERC, this PDG process occurs during the p/n junction formation. The gettering process can be included in addition to the cleaning and texturing steps of the SHJ fabrication process. Zhu et al. studied PDG for commercial n-type CZ silicon wafers for the application in SHJ solar cells. They demonstrated that PDG clearly improves the minority carrier lifetime of n-type CZ wafers and the performance of SHJ solar cells.[135]

For mitigating the losses resulting from oxygen, the quality of wafers should be upgraded. In the wafer specifications, oxygen concentration is an important parameter. The typical oxygen concentration of Cz-monocrystalline silicon wafers is ≈5–10 × 10^{17} cm^3. As the oxygen impurities originate mainly from the crucible during the ingot-pulling, therefore ingot quality could be improved by tighter purity control in the crucible, as well as tighter quality control of the raw silicon material.[50] For SHJ solar cells, besides reducing bulk defect density by gettering, the bulk SRH contribution to the losses can be mitigated by reducing wafer thickness.[136] Shorter diffusion lengths and thinner solar cells improve carrier collection and increase defect tolerance.[137] Approaches for higher open-circuit voltage are summarized in **Table 3**.

3.3. Roadmap toward Higher *FF*

The achieved maximum *FF* by LONGi (86.59%) is still lower than the theoretical FF limit (89.26%). *FF* losses in solar cells

Table 3. Summary of approaches for higher open-circuit voltage.

Approach for higher V_{OC}	
	Multiple H_2-plasma treatment method[119]
Suppression of recombination at surface	Dual-intrinsic layer/bilayer[120–122]
	Two-step method[13,129]
Suppression of recombination in bulk defects	Gettering[135]
Thinner silicon wafer	Thin wafer SHJ solar cells[111,112,137]
Post-treatment	Light-soaking[138]

come from series resistance in carrier transport R_{series} and carrier recombination.[139 140] As the solution to suppress carrier recombination has been discussed in earlier sections, in this section the solutions addressing the R_{series} will be discussed.

R_{series} in SHJ solar cells is related to various carrier transport mechanisms: the bulk resistance for the vertical free carriers' migration across the TCO layers and silicon-based layers, the sheet resistance for the in-plane free carriers' migration along with the TCO layers, the contact resistance, and the resistance of the metal electrodes.[75] Among these components, since the bulk resistance of TCO layers and grid resistance are mostly associated with the EQE and the J_{SC} of SHJ solar cells rather than *FF*, they contribute less to the FF loss.[54,56,58,141] In this case, the contact resistance is more important due to their major contributions to the total device R_{series}.[142–144] In 2016, Lachenal et al. analyzed the series resistance (≈0.62 Ω cm^2) of a high-efficiency bifacial rear-emitter HJT cell. According to their analysis, p-type a-Si:H /TCO contact is the major contribution of the R_{series} covering ≈38%, followed by the n-type a-Si:H/TCO contact covering ≈22%.

3.3.1. Reduction of Contact Resistance

The typical hole transport layer of SHJ solar cells is c-Si(n)/a-Si:H(p)/TCO(n), where tunneling across the junction heterointerfaces is the critical transport mechanism. The Schottky-like behavior of this contact is therefore governed by the band alignment at the interface and by the energy band diagram of the a-Si:H and TCO layer.[145] The preferred band diagram near the interface is an accumulating condition. According to the simulation by M. Bivour et al. in 2012, low doping concentration and high work function of p-type a-Si:H hole selective layers can induce the unwanted depletion or inversion of p-type to n-type hole selective layers. As a result, the doping concentration was increased.

Regarding the TCO layers, superior electrical and optical performances were optimized and balanced. Its work function has a strong influence on the contact resistance according to research by computer simulation.[146] Its optimal value is different on a-Si:H(n) BSF and a-Si:H(p) emitters and should be taken into consideration when designing a cell. In 2014, Ritzau et al. reported that work function engineering can improve contact.[147] According to their simulation, the work function of the hole contact should be high to match the Fermi level of the p-type a-Si:H (5.3 eV) or its valence band (5.6 eV). Based on these theories, when using ITO film, the mass ratio of In_2O_3/SnO_2 should be varied to optimize its work function. It was found that the 90/10-ITO (mass ratio of In_2O_3/SnO_2 = 90/10) layer has a lower work function than a 97/3-ITO (mass ratio of In_2O_3/SnO_2 = 97/3) layer and has better electronic matching with a-Si:H(n) layers.[148] The front side TCO film often fails to form an ideal electrical contact with the neighboring layer because its transmittance needs to be considered.[149] An alternative way to decrease the contact resistance is by implementing a buffer TCO layer, which could result in lowered series resistance and boost *FF* by 1%.[142]

Regarding p-type a-Si:H, a higher doping concentration is required to optimize the contact resistance by avoiding the hole selective layer from being depleted. Realizing this strategy to implement hydrogenated nanocrystalline (nc-Si:H) layers to the SHJ cell structure is now drawing attention.[69] Under thermal equilibrium, the band bending at c-Si(n)/a-Si:H(p) layer interface is indicated by the V_{bi} and it is defined as Equation (13):[80]

$$V_{bi,p} = \chi_{e,p} + E_{g,p} - E_{a,p} - \varphi_n \tag{13}$$

As analyzed by Procel et al. in 2018, low activation energy (E_a) and wide bandgap (E_g) are essential for (p)-contacts to achieve effective hole collection in silicon heterojunction (SHJ) solar cells.[80] For silicon-based films, the bandgap E_g can be modified by alloying carbon or oxygen with silicon. In 2021, Zhao et al. proposed a bi-layer (p)-contact structure to maximize the V_{bi}, consisting of p-type nc-SiO$_x$:H main layers and p-type nc-Si:H seed layers (as shown in **Figure 9**). To accelerate the nucleation of the nc-SiO$_x$:H layer, they additionally applied a combined interface treatment including hydrogen plasma treatment (HPT) and very-high-frequency (VHF) nc-Si:H(i) treatment before the (p)-contact deposition. A hole selective contact stack featuring a low contact resistance (ρ_c of 0.144 Ω cm^2) was achieved when a-Si:H(i) is 5 nm.[75]

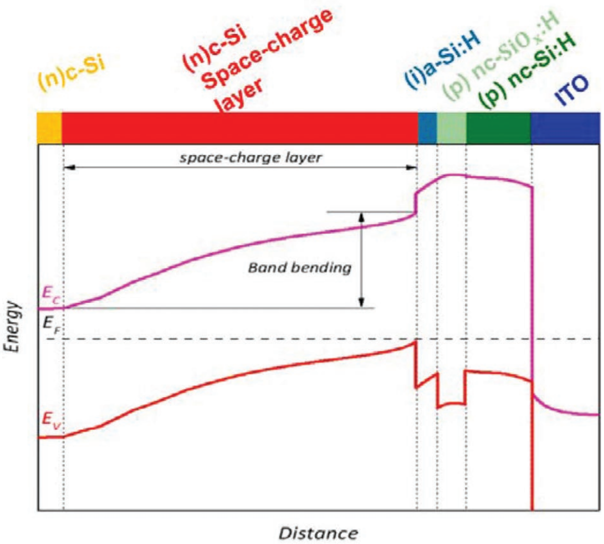

Figure 9. Schematic sketch (top) and band diagram under the dark thermal equilibrium condition (diagram) of the (p)-contact stack for SHJ solar cells with the bi-layer (p)-contact proposed by Zhao. The (p)-contact stack consists of a space-charge layer inside the (n)c-Si bulk, a-Si:H(i), nc-SiO$_x$:H(p), nc-Si:H(p), ITO. Reproduced with permission.[75] Copyright 2021, Elsevier.

4. Roadmap toward Industrialization of SHJ Solar Cells

Since 2003, Sanyo Corp. (now Panasonic Corp.) has been the first company to commercialize SHJ solar cells. Since then, this high-efficiency solar cell technology has attracted an increasing number of new PV players. In the last few years, several companies have launched mass production of SHJ solar cells, such as Gold Stone Solar (Fujian, China), HuaSun (Anhui, China), and REC Solar (Norway). SHJ solar cells are expected to gain a market share of ≈10% in 2025 and ≈18% by 2031.[150] Aside from PCE, the cost is a prime obstacle for industrialization. The total manufacturing costs of SHJ solar cells are ≈0.16 US\$ W^{-1} in 2020, which is higher than that of PERC solar cells. As shown in **Figure 10**, a large proportion of manufacturing costs originates from used materials, including silicon wafers and consumable materials. Benefiting from the manufacturing process with fewer steps, the human cost of SHJ solar cells is lower than PERC solar cells, contributing 2% of manufacturing costs. About 17% of costs are related to equipment, power, and facility. Among consumable materials, silver paste is the main cost in SHJ cells production, followed by ITO coatings. Recently, the sustainability of silicon PV production is being increasingly discussed. This is a more important aspect for SHJ cells because the current SHJ technology depends more on rare/volatile metals like silver, indium, and bismuth. The global reserves of silver, indium, and bismuth are estimated at 560, 50, and 320 kilotons. The global supply levels of silver, indium, and bismuth were only 29, 2.1, and 21 kilotons in 2019.[151] For long-term development on a terawatt scale, there must be a significant reduction in the usage of these scarce materials. Fortunately, a number of promising solutions for this concern are currently in the developmental and research stage.

4.1. Low Cost Metal Electrodes

In 2021, the price of Ag paste is ≈1100 US\$ Kg^{-1}. For SHJ solar cells of M6 wafers, ≈200 mg of Ag paste is needed for metallization. To reduce the need for silver, copper is the first choice because it not only exhibits similar electrical resistivity to silver but is also 100 times cheaper. In 2018, NAMICS Corp. developed low-temperature curing Ag coated Cu paste as an alternative approach.[152] This paste is made of Cu particles coated with Ag, which can drastically reduce the cost of low-temperature Ag paste. Cost reduction and conductor resistance depend on the Ag particle blend ratio in the Ag coated Cu paste (price of Ag coated Cu paste \$ Kg^{-1} = Ag particle blend ratio (%) × 1100 + 100). However, potential investors want to fully replace the consumption of Ag with Cu. Investigations into Cu contacts as an alternative to Ag contacts for SHJ solar cells have been mainly carried out by the method of electroplating, due to its various advantages, such as high aspect ratio and low contact resistance.[30] Cu plating for SHJ cells requires additional masking steps to pattern the TCO layer or a metal seed layer.[109] These masking steps and subsequent mask-removal steps add to cell processing costs and the complexity of the SHJ cells. For future mass production, the challenge is that the copper does not adhere well to the surface of SHJ solar cells.[103,104,153] The application of screen-printable copper paste on solar cells has been studied as it can be easily applied to established cell production lines. Copper paste for low-temperature curing has been researched as a promising product by some groups such as Dow Corning, AIST Japan, Samsung, et al.[154,155] The main issues of developing copper paste are the prevention of copper oxidation during annealing and the diffusion into the silicon substrate. Copper paste applications in SHJ solar cells have higher potential because the TCO layer acts as a diffusion barrier for copper at low curing temperatures.

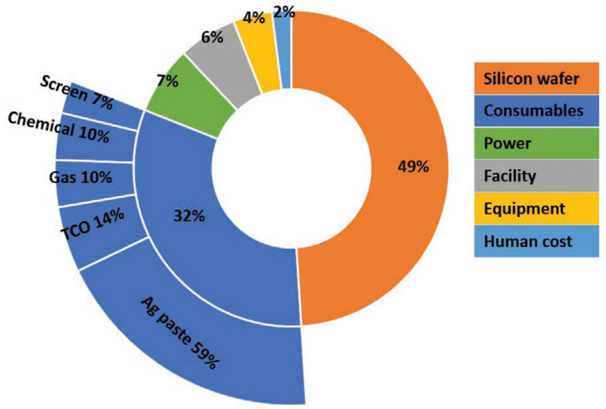

Figure 10. Breakdown of manufacturing cost of SHJ solar cells, total manufacturing costs of SHJ cells are ≈0.16 US\$ W^{-1}.[167]

4.2. Lower TCO Cost

ITO targets are another cost factor. Typically 60 to 80 nm thickness of TCO is deposited at the front and back of the SHJ solar cells, corresponding typically to usage of ≈3.5 grams of Indium per meter square of modules for both surfaces (taking into account a 50% usage of the sputtered materials).[59] Indium-free TCO layers are one research focus. Costs induced by indium oxide can be decreased by other substitute materials such as aluminum-doped zinc oxide (AZO).[156,157] However, the relatively poor conductivity and long-term stability of aluminum-doped zinc oxide (AZO) layers need to be solved.[158,159] Another approach to decrease costs induced by indium is to implement thinner ITO.[93,97] The implementation of a thinner TCO layer, however, requires a second layer on top, for example, SiO_2 or Si_3N_4 to maintain the anti-reflection (AR) optimum.[160,161] For sustainable manufacturing of SHJ solar cells, this approach seems to be a temporary solution.

4.3. Perovskite–Silicon Tandem Solar Cells to Lower LOCE

Next-generation PV devices combine low costs and high efficiency. Increasing the PCE of solar cells is one key driver to further decrease the LCOE. As efficiencies of current mainstream technologies approach theoretical limits, novel technologies are needed to continue to improve efficiency and drive down the cost of solar energy production.[162] A tandem solar cell, consisting of a silicon solar cell overlaid by a perovskite solar cell,

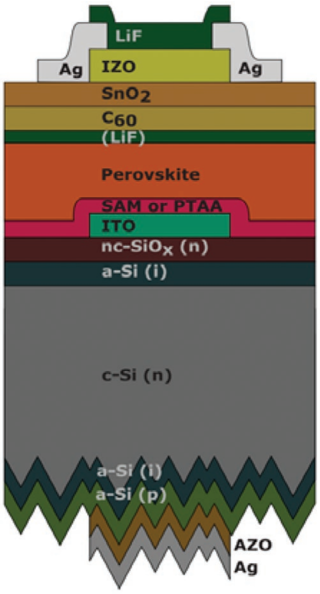

Figure 11. Schematic stack of the monolithic perovskite/SHJ tandem solar cell. Reproduced with permission.[163] Copyright 2020, AAAS.

could increase efficiencies of commercially mass-produced photovoltaics beyond the single-junction efficiency limit without adding substantial costs.[163] By carefully tuning the bandgap of the perovskite absorber, efficiency limits for perovskite-silicon solar cells are predicted to be 39%.[164] Perovskite–silicon tandem solar cells have shown rapid progress within the past five years in terms of their cell efficiency and are currently being investigated as candidates for the next generation of industrial PV devices.[165] Currently, silicon heterojunction (SHJ) bottom cell technology, which provides high open-circuit voltages, dominates tandem solar cell research, and is also achieving the highest efficiency levels in perovskite-silicon tandem solar cells (29.8% efficiency achieved by HZB), as shown in **Figure 11**.[163] The perovskite absorber of this high-efficiency tandem solar cell was spin-coated on the polished front side of a SHJ bottom cell.[163] Spin coating, which is often used in the laboratory, can be replaced by screen printing for large-scale manufacturing. Some researchers have performed detailed cost analyses on perovskite-silicon tandem modules. They have found that perovskite PV modules (both single junction and multi-junction) are competitive in the context of LCOE if the module lifespan is comparable with that of c-Si solar cells.[166] The high efficiency of perovskite/SHJ tandem solar cells yields a lower LCOE in comparison with the conventional PERC single junctions, in the order of ≈11%.[165] Therefore, perovskite-SHJ tandem solar cells can be seen as another promising approach to decrease the LOCE in future mass production.

5. Future Developments

PV technology is ready to become one of our main energy sources and contribute to carbon neutrality by the mid-21st century. We are close to the era of TW installed-scale PV. To meet the continued demand for high-efficiency solar cells, expectations for large-scale mass production of SHJ solar cells are rising. We have reviewed recent solutions for approaching the efficiency limit of silicon solar cells and the industrialization of SHJ solar cells. Regarding efficiency, in our point of view, the research focus should be to increase the short-circuit current while maintaining its high open-circuit voltage. For near-future mass production of SHJ solar cells, minimizing the consumption of silver and indium is essential. For long-term development, high-efficiency perovskite-SHJ tandem solar cells will likely take over.

Acknowledgements

The authors thank Hua Wu, Shi Yin, and Yong Liu for proofreading and fruitful discussions. Thanks to Shenghou Zhou for drawing. This work was supported by the National Natural Science Foundation of China (No. 61922005, 62034001, 61974008), the Beijing Municipal Natural Science Foundation (Grant No. JQ20027), the National Key Research and Development Program of China (Grant No. 2021YFA0715600).

Conflict of Interest

The authors declare no conflict of interest.

Keywords

heterojunctions, high-efficiency, industrialization, photovoltaics, solar cells

Received: January 3, 2022
Revised: March 17, 2022
Published online: April 10, 2022

[1] W. Shockley, H. J. Queisser, *J. Appl. Phys.* **1961**, *32*, 510.
[2] T. Tiedje, E. Yablonovitch, G. D. Cody, B. G. Brooks, *IEEE Trans. Electron Devices* **1984**, *31*, 711.
[3] A. Richter, M. Hermle, S. W. Glunz, *IEEE J. Photovoltaics* **2013**, *3*, 1184.
[4] S. Schafer, R. Brendel, *IEEE J. Photovoltaics* **2018**, *8*, 1156.
[5] J. Schmidt, R. Peibst, R. Brendel, *Sol. Energy Mater. Sol. Cells* **2018**, *187*, 39.
[6] W. Long, S. Yin, F. Peng, M. Yang, L. Fang, X. Ru, M. Qu, H. Lin, X. Xu, *Sol. Energy Mater. Sol. Cells* **2021**, *231*, 111291.
[7] A. Richter, J. Benick, F. Feldmann, A. Fell, B. Steinhauser, J.-I. Polzin, N. Tucher, J. N. Murthy, M. Hermle, S. W. Glunz, presented at 36th Eur. PV Sol. Energy Conf. Exhib., Marseille, September **2019**.
[8] A. Cuevas, *Energy Procedia* **2014**, *55*, 53.
[9] F. Feldmann, M. Bivour, C. Reichel, M. Hermle, S. W. Glunz, presented at 28th Eur. PV Sol. Energy Conf. Exhib., Paris, September **2013**.
[10] J. Melskens, B. W. H. van de Loo, B. Macco, L. E. Black, S. Smit, W. M. M. Kessels, *IEEE J. Photovoltaics* **2018**, *8*, 373.
[11] K. Wakisaka, M. Taguchi, T. Sawada, M. Tanaka, T. Matsuyama, T. Matsuoka, S. Tsuda, S. Nakano, Y. Kishi, Y. Kuwano, presented at Conf. Rec. Twenty-Second IEEE Photovoltaic Spec. Conf., Las Vegas, October **1991**.
[12] J. Geissbühler, J. Werner, S. Martin de Nicolas, L. Barraud, A. Hessler-Wyser, M. Despeisse, S. Nicolay, A. Tomasi, B. Niesen, S. De Wolf, C. Ballif, *Appl. Phys. Lett.* **2015**, *107*, 081601.
[13] X. Ru, M. Qu, J. Wang, T. Ruan, M. Yang, F. Peng, W. Long, K. Zheng, H. Yan, X. Xu, *Sol. Energy Mater. Sol. Cells* **2020**, *215*, 110643.

[14] M. A. Green, E. D. Dunlop, J. Hohl-Ebinger, M. Yoshita, N. Kopidakis, X. Hao, *Prog. Photovoltaics* **2021**, *30*, 3.
[15] K. Masuko, M. Shigematsu, T. Hashiguchi, D. Fujishima, M. Kai, N. Yoshimura, T. Yamaguchi, Y. Ichihashi, T. Mishima, N. Matsubara, T. Yamanishi, T. Takahama, M. Taguchi, E. Maruyama, S. Okamoto, *IEEE J. Photovoltaics* **2014**, *4*, 1433.
[16] K. Yoshikawa, W. Yoshida, T. Irie, H. Kawasaki, K. Konishi, H. Ishibashi, T. Asatani, D. Adachi, M. Kanematsu, H. Uzu, K. Yamamoto, *Sol. Energy Mater. Sol. Cells* **2017**, *173*, 37.
[17] K. Yoshikawa, H. Kawasaki, W. Yoshida, T. Irie, K. Konishi, K. Nakano, T. Uto, D. Adachi, M. Kanematsu, H. Uzu, K. Yamamoto, *Nat. Energy* **2017**, *2*, 17032.
[18] I. M. Peters, C. D. R. Gallegos, S. E. Sofia, T. Buonassisi, *Joule* **2019**, *3*, 2732.
[19] J. Haschke, J. P. Seif, Y. Riesen, A. Tomasi, J. Cattin, L. Tous, P. Choulat, M. Aleman, E. Cornagliotti, A. Uruena, R. Russell, F. Duerinckx, J. Champliaud, J. Levrat, A. A. Abdallah, B. Aïssa, N. Tabet, N. Wyrsch, M. Despeisse, J. Szlufcik, S. De Wolf, C. Ballif, *Energy Environ. Sci.* **2017**, *10*, 1196.
[20] C. Ballif, M. Boccard, M. Despeisse, presented at 7th World Conf. Photovoltaic Energy Convers., Hawaii, June **2018**.
[21] T. S. Liang, M. Pravettoni, C. Deline, J. S. Stein, R. Kopecek, J. P. Singh, W. Luo, Y. Wang, A. G. Aberle, Y. S. Khoo, *Energy Environ. Sci.* **2019**, *12*, 116.
[22] S. Alivizatos, *Master Thesis*, Dlelft University of Technology, **2013**.
[23] S. De Wolf, A. Descoeudres, Z. C. Holman, C. Ballif, *Green* **2012**, *2*, 7.
[24] K. Higashimine, K. Koyama, K. Ohdaira, H. Matsumura, N. Otsuka, *J. Vac. Sci. Technol., B: Nanotechnol. Microelectron.: Mater., Process., Meas., Phenom.* **2012**, *30*, 031208.
[25] D. Yan, A. Cuevas, J. I. Michel, C. Zhang, Y. Wan, X. Zhang, J. Bullock, *Joule* **2021**, *5*, 811.
[26] M. Taguchi, K. Kawamoto, S. Tsuge, T. Baba, H. Sakata, M. Morizane, K. Uchihashi, N. Nakamura, S. Kiyama, O. Oota, *Prog. Photovoltaics* **2000**, *8*, 503.
[27] S. O. M. Tanaka, S. Okamoto, S. Tsuge, S. Kiyama, presented at 3rd World Conf. Photovoltaic Energy Convers., Osaka, May **2003**.
[28] Y. T. M. Taguchi, H. Inoue, S. Taira, T. Nakashima, T. Baba, H. Sakata, E. Maruyama, presented at 24th Eur. PV Sol. Energy Conf. Exhib., Hamburg, September **2009**.
[29] D. F. T. Kinoshita, A. Yano, A. Ogane, S. Tohoda, K. Matsuyama, Y. Nakamura, N. Tokuoka, H. Kanno, H. Sakata, M. Taguchi, E. Maruyama, presented at 26th Eur. PV Sol. Energy Conf. Exhib., Hamburg, September **2011**.
[30] J. L. Hernández, D. Adachi, D. Schroos, N. Valckx, N. Menou, T. Uto, M. Hino, M. Kanematsu, H. Kawasaki, R. Mishima, K. Nakano, H. Uzu, T. Terashita, K. Yoshikawa, T. Kuchiyama, M. Hiraishi, N. Nakanishi, M. Yoshimi, K. Yamamoto, presented at 28th Eur. PV Sol. Energy Conf. Exhib., Paris, September **2013**.
[31] K. H. Kim, C. S. Park, J. D. Lee, J. Y. Lim, J. M. Yeon, I. H. Kim, E. J. Lee, Y. H. Cho, *Jpn. J. Appl. Phys.* **2017**, *56*, 08MB25.
[32] M. A. Green, *Prog. Photovoltaics* **2009**, *17*, 183.
[33] A. Richter, R. Müller, J. Benick, F. Feldmann, B. Steinhauser, C. Reichel, A. Fell, M. Bivour, M. Hermle, S. W. Glunz, *Nat. Energy* **2021**, *6*, 429.
[34] A. Zangwill, *Physics at Surfaces*, Cambridge University Press, Cambridge **1988**.
[35] W. G. Van Sark, L. Korte, F. Roca, *Physics and Technology of Amorphous-Crystalline Heterostructure Silicon Solar Cells*, Springer, Berlin, Germany **2012**.
[36] J. R. Ligenza, *J. Phys. Chem.* **2002**, *65*, 2011.
[37] D. Adachi, J. L. Hernández, K. Yamamoto, *Appl. Phys. Lett.* **2015**, *107*, 233506.
[38] K. R. McIntosh, L. E. Black, *J. Appl. Phys.* **2014**, *116*, 014503.
[39] S. Y. Herasimenka, W. J. Dauksher, S. G. Bowden, *Appl. Phys. Lett.* **2013**, *103*, 053511.
[40] A. Descoeudres, Z. C. Holman, L. Barraud, S. Morel, S. De Wolf, C. Ballif, *IEEE J. Photovoltaics* **2013**, *3*, 83.
[41] Z. C. Holman, M. Filipič, A. Descoeudres, S. De Wolf, F. Smole, M. Topič, C. Ballif, *J. Appl. Phys.* **2013**, *113*, 013107.
[42] A. Liu, S. P. Phang, D. Macdonald, *Sol. Energy Mater. Sol. Cells* **2022**, *234*, 111447.
[43] J. R. Davis, A. Rohatgi, R. H. Hopkins, P. D. Blais, P. Rai-Choudhury, J. R. McCormick, H. C. Mollenkopf, *IEEE Trans. Electron Devices* **1980**, *27*, 677.
[44] T. Buonassisi, A. A. Istratov, M. D. Pickett, M. Heuer, J. P. Kalejs, G. Hahn, M. A. Marcus, B. Lai, Z. Cai, S. M. Heald, T. F. Ciszek, R. F. Clark, D. W. Cunningham, A. M. Gabor, R. Jonczyk, S. Narayanan, E. Sauar, E. R. Weber, *Prog. Photovoltaics* **2006**, *14*, 513.
[45] A. A. Istratov, T. Buonassisi, R. J. McDonald, A. R. Smith, R. Schindler, J. A. Rand, J. P. Kalejs, E. R. Weber, *J. Appl. Phys.* **2003**, *94*, 6552.
[46] B. Aissa, M. Kivambe, M. Hossain, O. Daif, A. Abdallah, F. Ali, N. Tabet, *Front. Nanosci. Nanotechnol.* **2015**, *1*, 2.
[47] S. W. Glunz, S. Rein, J. Y. Lee, W. Warta, *J. Appl. Phys.* **2001**, *90*, 2397.
[48] M. Tomassini, J. Veirman, R. Varache, E. Letty, S. Dubois, Y. Hu, Ø. Nielsen, *J. Appl. Phys.* **2016**, *119*, 084508.
[49] F. Meng, presented at *3rd Int. Workshop on Silicon Heterojunction Sol. Cells: Sci. Ind. Technol.*, Virtual workshop, October **2020**.
[50] J. Zhao, M. Konig, Y. Yao, Y. C. Wang, R. Zhou, T. Xie, H. Deng, presented at 7th World Conf. Photovoltaic Energy Convers., Hawaii, June **2018**.
[51] T. G. Allen, J. Bullock, X. Yang, A. Javey, S. De Wolf, *Nat. Energy* **2019**, *4*, 914.
[52] A. Descoeudres, J. Horzel, B. Paviet-Salomon, L. L. Senaud, G. Christmann, J. Geissbühler, P. Wyss, N. Badel, J. W. Schüttauf, J. Zhao, C. Allebé, A. Faes, S. Nicolay, C. Ballif, M. Despeisse, *Prog. Photovoltaics* **2019**, *28*, 569.
[53] M. Hermle, F. Feldmann, M. Bivour, J. C. Goldschmidt, S. W. Glunz, *Appl. Phys. Rev.* **2020**, *7*, 021305.
[54] A. Valla, P. Carroy, F. Ozanne, D. Muñoz, *Sol. Energy Mater. Sol. Cells* **2016**, *157*, 874.
[55] F. Meng, J. Shi, Z. Liu, Y. Cui, Z. Lu, Z. Feng, *Sol. Energy Mater. Sol. Cells* **2014**, *122*, 70.
[56] Y. He, G. Dong, G. Cui, H. Lu, T. Lan, Z. Shen, W. Long, C. Yu, H. Yan, J. Zhang, Y. Li, X. Xu, presented at 7th World Conf. Photovoltaic Energy Convers., Hawaii, June **2018**.
[57] R. Kopecek, J. Libal, *Nat. Energy* **2018**, *3*, 443.
[58] S. N. A. Cruz, A. B. Morales, R. Schlatmann, presented at 35th Eur. PV Sol. Energy Conf. Exhib., Brussels, September **2018**.
[59] J. Haschke, O. Dupré, M. Boccard, C. Ballif, *Sol. Energy Mater. Sol. Cells* **2018**, *187*, 140.
[60] Z. C. Holman, A. Descoeudres, S. De Wolf, C. Ballif, *IEEE J. Photovoltaics* **2013**, *3*, 1243.
[61] R. A. Sinton, A. Cuevas, presented at 16th Eur. PV Sol. Energy Conf. Exhib., Glasgow, May **2000**.
[62] R. Brendel, S. Dreissigacker, N. P. Harder, P. P. Altermatt, *Appl. Phys. Lett.* **2008**, *93*, 173503.
[63] N. Brinkmann, G. Micard, Y. Schiele, G. Hahn, B. Terheiden, *Phys. Status Solidi RRL* **2013**, *7*, 322.
[64] P. Saint-Cast, S. Werner, J. Greulich, U. Jäger, E. Lohmüller, H. Höffler, R. Preu, *Phys. Status Solidi A* **2017**, *214*, 1600708.
[65] Z. C. Holman, A. Descoeudres, L. Barraud, F. Z. Fernandez, J. P. Seif, S. De Wolf, C. Ballif, *IEEE J. Photovoltaics* **2012**, *2*, 7.
[66] A. Paduthol, M. K. Juhl, G. Nogay, P. Löper, T. Trupke, *Prog. Photovoltaics* **2018**, *26*, 968.
[67] H. Kang, *IOP Conf. Ser.: Earth Environ. Sci.* **2021**, *726*, 012001.
[68] L. Mazzarella, S. Kirner, O. Gabriel, S. S. Schmidt, L. Korte, B. Stannowski, B. Rech, R. Schlatmann, *Phys. Status Solidi A* **2017**, *214*, 1532958.
[69] J. P. Seif, A. Descoeudres, G. Nogay, S. Hanni, S. M. de Nicolas, N. Holm, J. Geissbuhler, A. Hessler-Wyser, M. Duchamp,

R. E. Dunin-Borkowski, M. Ledinsky, S. De Wolf, C. Ballif, *IEEE J. Photovoltaics* **2016**, *6*, 1132.
[70] D. Zhang, D. Deligiannis, G. Papakonstantinou, R. A. C. M. M. van Swaaij, M. Zeman, *IEEE J. Photovoltaics* **2014**, *4*, 1326.
[71] Y. Zhang, C. Yu, M. Yang, Y. He, L. Zhang, J. Zhang, X. Xu, Y. Zhang, X. Song, H. Yan, *RSC Adv.* **2017**, *7*, 9258.
[72] M. Köhler, M. Pomaska, P. Procel, R. Santbergen, A. Zamchiy, B. Macco, A. Lambertz, W. Duan, P. Cao, B. Klingebiel, S. Li, A. Eberst, M. Luysberg, K. Qiu, O. Isabella, F. Finger, T. Kirchartz, U. Rau, K. Ding, *Nat. Energy* **2021**, *6*, 529.
[73] H. Fujiwara, T. Kaneko, M. Kondo, *Appl. Phys. Lett.* **2007**, *91*, 133508.
[74] J. Peter Seif, A. Descoeudres, M. Filipič, F. Smole, M. Topič, Z. Charles Holman, S. De Wolf, C. Ballif, *J. Appl. Phys.* **2014**, *115*, 024502.
[75] Y. Zhao, P. Procel, C. Han, L. Mazzarella, G. Yang, A. Weeber, M. Zeman, O. Isabella, *Sol. Energy Mater. Sol. Cells* **2021**, *219*, 110779.
[76] Y. Zhao, L. Mazzarella, P. Procel, C. Han, G. Yang, A. Weeber, M. Zeman, O. Isabella, *Prog. Photovoltaics* **2020**, *28*, 425.
[77] L. Mazzarella, A. B. Morales-Vilches, M. Hendrichs, S. Kirner, L. Korte, R. Schlatmann, B. Stannowski, *IEEE J. Photovoltaics* **2018**, *8*, 70.
[78] J. Haschke, R. Monnard, L. Antognini, J. Cattin, A. A. Abdallah, B. Aïssa, M. M. Kivambe, N. Tabet, M. Boccard, C. Ballif, *AIP Conf. Proc.* **2018**, *1999*, 030001.
[79] C.-W. Peng, C. Lei, T. Ruan, J. Zhong, M. Yang, W. Long, C. Yu, Y. Li, X. Xu, presented at IEEE 46th Photovoltaic Spec. Conf., Chicago, June **2019**.
[80] P. Procel, G. Yang, O. Isabella, M. Zeman, *Sol. Energy Mater. Sol. Cells* **2018**, *186*, 66.
[81] A. Cruz, D. Erfurt, R. Köhler, M. Dimer, E. Schneiderlöchner, B. Stannowski, *Photovoltaics Int.* **2020**, *44*, 86.
[82] H. Li, S. Yin, G. Dong, G. Cui, M. Chen, J. Chen, X. Xu, L. Feng, J. Zhang, C. Yu, *IEEE J. Photovoltaics* **2021**, *11*, 312.
[83] M. Balestrieri, D. Pysch, J. P. Becker, M. Hermle, W. Warta, S. W. Glunz, *Sol. Energy Mater. Sol. Cells* **2011**, *95*, 2390.
[84] J. Yu, J. Bian, W. Duan, Y. Liu, J. Shi, F. Meng, Z. Liu, *Sol. Energy Mater. Sol. Cells* **2016**, *144*, 359.
[85] Z. Lu, F. Meng, Y. Cui, J. Shi, Z. Feng, Z. Liu, *J. Phys. D: Appl. Phys.* **2013**, *46*, 075103.
[86] T. Koida, H. Fujiwara, M. Kondo, *Jpn. J. Appl. Phys.* **2007**, *46*, L685.
[87] T. Koida, M. Kondo, K. Tsutsumi, A. Sakaguchi, M. Suzuki, H. Fujiwara, *J. Appl. Phys.* **2010**, *107*, 033514.
[88] L. Tutsch, H. Sai, T. Matsui, M. Bivour, M. Hermle, T. Koida, *Prog. Photovoltaics* **2021**, *29*, 835.
[89] E. Kobayashi, Y. Watabe, T. Yamamoto, *Appl. Phys. Express* **2015**, *8*, 015505.
[90] Z. Yao, W. Duan, P. Steuter, J. Hüpkes, A. Lambertz, K. Bittkau, M. Pomaska, D. Qiu, K. Qiu, Z. Wu, H. Shen, U. Rau, K. Ding, *Sol. RRL* **2021**, *5*, 2000501.
[91] M. Morales-Masis, E. Rucavado, R. Monnard, L. Barraud, J. Holovsky, M. Despeisse, M. Boccard, C. Ballif, *IEEE J. Photovoltaics* **2018**, *8*, 1202.
[92] T. Koida, M. Kondo, *J. Appl. Phys.* **2007**, *101*, 063713.
[93] A. Cruz, E.-C. Wang, A. B. Morales-Vilches, D. Meza, S. Neubert, B. Szyszka, R. Schlatmann, B. Stannowski, *Sol. Energy Mater. Sol. Cells* **2019**, *195*, 339.
[94] M. Bivour, S. Schröer, M. Hermle, S. W. Glunz, *Sol. Energy Mater. Sol. Cells* **2014**, *122*, 120.
[95] M. Bivour, H. Steinkemper, J. Jeurink, S. Schröer, M. Hermle, *Energy Procedia* **2014**, *55*, 229.
[96] S. Li, M. Pomaska, A. Lambertz, W. Duan, K. Bittkau, D. Qiu, Z. Yao, M. Luysberg, P. Steuter, M. Köhler, K. Qiu, R. Hong, H. Shen, F. Finger, T. Kirchartz, U. Rau, K. Ding, *Joule* **2021**, *5*, 1535.
[97] P. P. D. L. Bätzner, B. Legradic, D. Lachenal, R. Kramer, T. Kössler, L. Andreetta, N. Holm, W. Frammelsberger, B. Strahm, presented at 36th Eur. PV Sol. Energy Conf. Exhib., Marseille, September **2019**.
[98] J. Yu, J. Zhou, J. Bian, L. Zhang, Y. Liu, J. Shi, F. Meng, J. Liu, Z. Liu, *Jpn. J. Appl. Phys.* **2017**, *56*, 08MB09.
[99] S. Braun, G. Hahn, R. Nissler, C. Pönisch, D. Habermann, *Energy Procedia* **2013**, *43*, 86.
[100] S. Braun, G. Hahn, R. Nissler, C. Pönisch, D. Habermann, *Energy Procedia* **2013**, *38*, 334.
[101] P. Papet, L. Andreetta, D. Lachenal, G. Wahli, J. Meixenberger, B. Legradic, W. Frammelsberger, D. Bätzner, B. Strahm, Y. Yao, T. Söderström, *Energy Procedia* **2015**, *67*, 203.
[102] H. Schulte-Huxel, S. Blankemeyer, A. Morlier, R. Brendel, M. Köntges, *Sol. Energy Mater. Sol. Cells* **2019**, *200*, 109991.
[103] A. U. Rehman, S. H. Lee, *Materials* **2014**, *7*, 1318.
[104] J. Yu, J. Li, Y. Zhao, A. Lambertz, T. Chen, W. Duan, W. Liu, X. Yang, Y. Huang, K. Ding, *Sol. Energy Mater. Sol. Cells* **2021**, *224*, 110993.
[105] J. Chang, F.-S. Chen, M. Y. Chen, W.-C. Hsieh, M.-Y. Huang, presented at 29th Eur. PV Sol. Energy Conf. Exhib., Amsterdam, September **2014**.
[106] H. Huang, L. Zhou, J. Yuan, Z. Quan, *Appl. Phys.* **2019**, *1906*, 6010.
[107] H. Huang, L. Zhou, J. Yuan, Z. Quan, *Chin. Phys. B* **2019**, *28*, 128503.
[108] L. Tous, S. N. Granata, A. Rouhi, J.-F. Lerat, T. Emeraud, S. M. d. Nicolas, T. M. Pletzer, R. Labie, M. Aleman, R. Russell, J. John, F. Duerinckx, J. Szlufcik, S. De Wolf, C. Ballif, R. Mertens, J. Poortmans, *Energy Procedia* **2014**, *55*, 715.
[109] G. Limodio, Y. De Groot, G. Van Kuler, L. Mazzarella, Y. Zhao, P. Procel, G. Yang, O. Isabella, M. Zeman, *IEEE J. Photovoltaics* **2020**, *10*, 372.
[110] A. Augusto, S. Y. Herasimenka, R. R. King, S. G. Bowden, C. Honsberg, *J. Appl. Phys.* **2017**, *121*, 205704.
[111] H. Sai, T. Oku, Y. Sato, M. Tanabe, T. Matsui, K. Matsubara, *Prog. Photovoltaics* **2019**, *27*, 1061.
[112] H. Sai, H. Umishio, T. Matsui, *Sol. RRL* **2021**, *5*, 2100634.
[113] C. Ballif, S. De Wolf, A. Descoeudres, Z. C. Holman, in *Advances in Photovoltaics: Part 3*, Vol. *90*, Elsevier, San Diego, USA **2014**, Ch. 2.
[114] S. Olibet, E. Vallat-Sauvain, L. Fesquet, C. Monachon, A. Hessler-Wyser, J. Damon-Lacoste, S. De Wolf, C. Ballif, *Phys. Status Solidi A* **2010**, *207*, 651.
[115] T. H. Wang, *IEEE Trans. Electron Devices* **2005**, *51*, 1048.
[116] S. De Wolf, M. Kondo, *Appl. Phys. Lett.* **2007**, *90*, 042111.
[117] H. Fujiwara, M. Kondo, *Appl. Phys. Lett.* **2007**, *90*, 013503.
[118] A. Descoeudres, L. Barraud, R. Bartlome, G. Choong, S. De Wolf, F. Zicarelli, C. Ballif, *Appl. Phys. Lett.* **2010**, *97*, 183505.
[119] A. Descoeudres, L. Barraud, S. De Wolf, B. Strahm, D. Lachenal, C. Guérin, Z. C. Holman, F. Zicarelli, B. Demaurex, J. Seif, J. Holovsky, C. Ballif, *Appl. Phys. Lett.* **2011**, *99*, 123506.
[120] K. S. Lee, C. B. Yeon, S. J. Yun, K. H. Jung, J. W. Lim, *ECS Solid State Lett.* **2014**, *3*, P33.
[121] L. Bodlak, J. Temmler, A. Moldovan, J. Rentsch, *AIP Conf. Proc.* **2019**, *2147*, 050001.
[122] H. Sai, H.-J. Hsu, P.-W. Chen, P.-L. Chen, T. Matsui, *Phys. Status Solidi A* **2021**, *218*, 2000743.
[123] Y. Zhang, C. Yu, M. Yang, L.-R. Zhang, Y.-C. He, J.-Y. Zhang, X.-X. Xu, Y.-Z. Zhang, X.-M. Song, H. Yan, *Chin. Phys. Lett.* **2017**, *34*, 038101.
[124] T. Ruan, M. Qu, X. Qu, X. Ru, J. Wang, Y. He, K. Zheng, B. H. H. Lin, X. Xu, Y. Zhang, H. Yan, *Thin Solid Films* **2020**, *711*, 138305.
[125] A. B. Morales-Vilches, E.-C. Wang, T. Henschel, M. Kubicki, A. Cruz, S. Janke, L. Korte, R. Schlatmann, B. Stannowski, *Phys. Status Solidi A* **2019**, *217*, 1900518.
[126] H. Sai, P.-W. Chen, H.-J. Hsu, T. Matsui, S. Nunomura, K. Matsubara, *J. Appl. Phys.* **2018**, *124*, 103102.
[127] H. Matsumura, K. Higashimine, K. Koyama, K. Ohdaira, *J. Vac. Sci. Technol. B* **2015**, *33*, 031201.
[128] Y. Yan, M. Page, T. H. Wang, M. M. Al-Jassim, H. M. Branz, Q. Wang, *Appl. Phys. Lett.* **2006**, *88*, 121925.

[129] X. Qu, Y. He, M. Qu, T. Ruan, F. Chu, Z. Zheng, Y. Ma, Y. Chen, X. Ru, X. Xu, H. Yan, L. Wang, Y. Zhang, X. Hao, Z. Hameiri, Z.-G. Chen, L. Wang, K. Zheng, *Nat. Energy* **2021**, *6*, 194.
[130] S. M. Myers, M. Seibt, W. Schröter, *J. Appl. Phys.* **2000**, *88*, 3795.
[131] J. S. Kang, D. K. Schroder, *J. Appl. Phys.* **1989**, *65*, 2974.
[132] E. Ö. Sveinbjörnsson, O. Engström, U. Södervall, *J. Appl. Phys.* **1993**, *73*, 7311.
[133] E. Spiecker, M. Seibt, W. Schröter, *Phys. Rev. B* **1997**, *55*, 9577.
[134] H. Park, S. J. Tark, C. S. Kim, S. Park, Y. D. Kim, C.-S. Son, J. C. Lee, D. Kim, *Int. J. Photoenergy* **2012**, *2012*, 794876.
[135] F. Zhu, D. Wang, J. Bian, J. Liu, Z. Liu, *Sol. Energy Mater. Sol. Cells* **2016**, *157*, 74.
[136] A. Augusto, J. Karas, P. Balaji, S. G. Bowden, R. R. King, *J. Mater. Chem. A* **2020**, *8*, 16599.
[137] A. Augusto, E. Looney, C. del Cañizo, S. G. Bowden, T. Buonassisi, *Energy Procedia* **2017**, *124*, 706.
[138] E. Kobayashi, S. De Wolf, J. Levrat, A. Descoeudres, M. Despeisse, F.-J. Haug, C. Ballif, *Sol. Energy Mater. Sol. Cells* **2017**, *173*, 43.
[139] G. P. Willeke, E. R. Weber, *Advances in Photovoltaics: Part 3*, San Diego **2014**.
[140] A. Khanna, T. Mueller, R. A. Stangl, B. Hoex, P. K. Basu, A. G. Aberle, *IEEE J. Photovoltaics* **2013**, *3*, 1170.
[141] A. Lachowicz, G. Andreatta, N. Blondiaux, A. Faes, N. Badel, J. J. D. Leon, C. Allébe, C. Fontaine, P.-H. Haumesser, J. Jourdan, D. Muñoz, M. Godard, M. Darmon, S. Nicolay, M. Despeisse, C. Ballif, *AIP Conf. Proc.* **2021**, *2367*, 020010.
[142] G. Dong, S. Yin, G. Cui, C. Yu, W. Long, H. Li, J. Zhang, Y. Li, X. Xu, presented at 46th IEEE Photovoltaic Spec. Conf., Chicago, IL, June **2019**.
[143] C. Luderer, C. Messmer, M. Hermle, M. Bivour, *IEEE J. Photovoltaics* **2020**, *10*, 952.
[144] Z. Wu, W. Duan, A. Lambertz, D. Qiu, M. Pomaska, Z. Yao, U. Rau, L. Zhang, Z. Liu, K. Ding, *Appl. Surf. Sci.* **2021**, *542*, 148749.
[145] M. Bivour, C. Reichel, M. Hermle, S. W. Glunz, *Sol. Energy Mater. Sol. Cells* **2012**, *106*, 11.
[146] L. Zhao, C. L. Zhou, H. L. Li, H. W. Diao, W. J. Wang, *Phys. Status Solidi A* **2008**, *205*, 1215.
[147] K.-U. Ritzau, M. Bivour, S. Schröer, H. Steinkemper, P. Reinecke, F. Wagner, M. Hermle, *Sol. Energy Mater. Sol. Cells* **2014**, *131*, 9.
[148] F.-J. Haug, R. Biron, G. Kratzer, F. Leresche, J. Besuchet, C. Ballif, M. Dissel, S. Kretschmer, W. Soppe, P. Lippens, K. Leitner, *Prog. Photovoltaics* **2012**, *20*, 727.
[149] P. Procel, H. Xu, A. Saez, C. Ruiz-Tobon, L. Mazzarella, Y. Zhao, C. Han, G. Yang, M. Zeman, O. Isabella, *Prog. Photovoltaics* **2020**, *28*, 935.
[150] M. Fischer, M. Woodhouse, S. Herritsch, J. Trube, *International Technology Roadmap for Photovoltaic (ITRPV 2021 results)*, VDMA Photovoltaic Equipment, Frankfurt am Main, Germany **2021**.
[151] Y. Zhang, M. Kim, L. Wang, P. Verlinden, B. Hallam, *Energy Environ. Sci.* **2021**, *14*, 5587.
[152] K. M. Kyotaro Nakamura, A. Tanaka, Y. Ohshita, presented at 35th Eur. PV Sol. Energy Conf. Exhib., Brussels, September **2018**.
[153] D. CARROLL, Australian start-up sets 25.54% efficiency record for silicon cell, https://www.pv-magazine.com/2021/09/10/australian-startup-sets-25-54-efficiency-record-for-silicon-cell/ (accessed: September 2021).
[154] S. H. Lee, S. H. Lee, in *Recent Developments in Photovoltaic Materials and Devices*, IntechOpen, London, UK **2019**, Ch. 2.
[155] B. H. Teo, A. Khanna, V. Shanmugam, M. L. O. Aguilar, M. E. Delos Santos, D. J. W. Chua, W.-C. Chang, T. Mueller, *Sol. Energy* **2019**, *189*, 179.
[156] A. B. Morales-Vilches, A. Cruz, S. Pingel, S. Neubert, L. Mazzarella, D. Meza, L. Korte, R. Schlatmann, B. Stannowski, *IEEE J. Photovoltaics* **2019**, *9*, 34.
[157] L.-L. Senaud, G. Christmann, A. Descoeudres, J. Geissbuhler, L. Barraud, N. Badel, C. Allebe, S. Nicolay, M. Despeisse, B. Paviet-Salomon, C. Ballif, *IEEE J. Photovoltaics* **2019**, *9*, 1217.
[158] D. Meza, A. Cruz, A. Morales-Vilches, L. Korte, B. Stannowski, *Appl. Sci.* **2019**, *9*, 862.
[159] P.-H. Chen, W.-J. Chen, J.-Y. Tseng, *Coatings* **2021**, *11*, 1546.
[160] D. Zhang, I. A. Digdaya, R. Santbergen, R. A. C. M. M. van Swaaij, P. Bronsveld, M. Zeman, J. A. M. van Roosmalen, A. W. Weeber, *Sol. Energy Mater. Sol. Cells* **2013**, *117*, 132.
[161] S. Y. Herasimenka, W. J. Dauksher, M. Boccard, S. Bowden, *Sol. Energy Mater. Sol. Cells* **2016**, *158*, 98.
[162] S. E. Sofia, H. Wang, A. Bruno, J. L. Cruz-Campa, T. Buonassisi, I. M. Peters, *Sustainable Energy Fuels* **2020**, *4*, 852.
[163] A. Al-Ashouri, E. Kohnen, B. Li, A. Magomedov, H. Hempel, P. Caprioglio, J. A. Marquez, A. B. M. Vilches, E. Kasparavicius, J. A. Smith, N. Phung, D. Menzel, M. Grischek, L. Kegelmann, D. Skroblin, C. Gollwitzer, T. Malinauskas, M. Jost, G. Matic, B. Rech, R. Schlatmann, M. Topic, L. Korte, A. Abate, B. Stannowski, D. Neher, M. Stolterfoht, T. Unold, V. Getautis, S. Albrecht, *Science* **2020**, *370*, 1300.
[164] M. T. Hörantner, T. Leijtens, M. E. Ziffer, G. E. Eperon, M. G. Christoforo, M. D. McGehee, H. J. Snaith, *ACS Energy Lett.* **2017**, *2*, 2506.
[165] C. Messmer, B. S. Goraya, S. Nold, P. S. C. Schulze, V. Sittinger, J. Schön, J. C. Goldschmidt, M. Bivour, S. W. Glunz, M. Hermle, *Prog. Photovoltaics* **2020**, *29*, 744.
[166] Z. Li, Y. Zhao, X. Wang, Y. Sun, Z. Zhao, Y. Li, H. Zhou, Q. Chen, *Joule* **2018**, *2*, 1559.
[167] W. Wang, presented at 3rd Chin. SHJ, Perovskite Tandem Sol. Cells Forum, Suzhou, May **2021**.

Zhaoqing Sun received the Ingénieur degree from Ecole Nationale Supérieure d' Ingénieur de Limoges. He is currently a Ph.D. student in the Faculty of Materials and Manufacturing, Beijing University of Technology. Previously he was an R&D engineer of laser processing and material surface modification at companies. His research interests include the fundamental understanding of silicon heterojunction photovoltaic devices as well as the laser process for silicon solar cells.

Xiaoqing Chen got his Ph.D. degree from the Department of Physics of Fudan University in China. Since then, he consequently worked in Linköping University in Sweden and the National Institute for Materials Science (NIMS) in Japan. At present, Xiaoqing Chen works at Beijing University of Technology in China. His main interest is the characterization and analysis of the dynamic processes in photoelectronic devices including solar cells and photodetectors.

Yongzhe Zhang received a Ph.D. degree in Condensed Matter Physics from Lanzhou University. He is currently a Professor of Electronic Engineering and Information Technology at Beijing University of Technology. Previous he was a Research Scientist at the school of Electrical and Electronic Engineering of Nanyang Technological University in Singapore. His research interests focus on improving the performance of silicon heterojunction solar cells and semiconductor photoelectric detectors, with an emphasis on interface engineering.